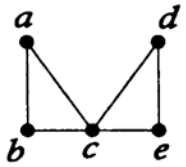
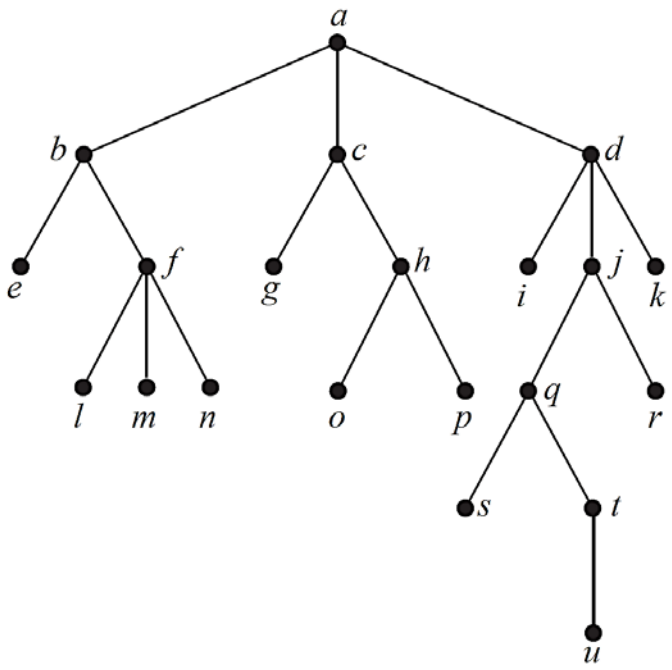
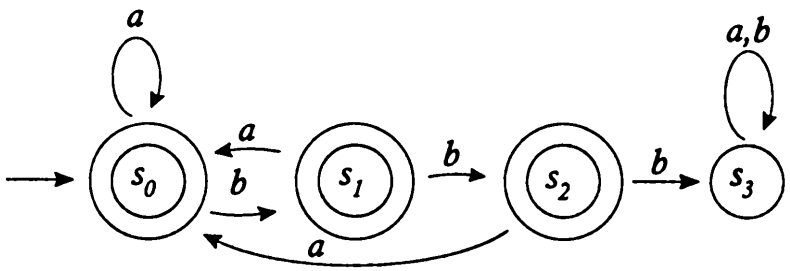
<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the converse of this conditional statement. If it snows today, I will sky tomorrow.</td><td>1. I will sky tomorrow only if it snows today.</td></tr>
<tr><td>2. If I do not sky tomorrow, then it will not have snowed today.</td></tr>
<tr><td>3. If it does not snow today, then I will not sky tomorrow.</td></tr>
<tr><td>4. If it snows today, then I will not sky tomorrow.</td></tr>
<tr><td>5. If it does not snow today, then I will sky tomorrow.</td></tr>
<tr><td rowspan="5">Assignment 2
A palindrome is a string of digits that reads the same backwards as forwards. How many different palindromes are there with six digits?</td><td>1. 900</td></tr>
<tr><td>2. 1000</td></tr>
<tr><td>3. 9000</td></tr>
<tr><td>4. 850</td></tr>
<tr><td>5. 10000</td></tr>
<tr><td rowspan="5">Assignment 3
Each of the x first-year students studying computing at Yanka Kupala Grodno State University can study a number of optional units. Of these, 25 chose to study the accounting option, 13 chose the business option and 8 studied both of these options/ Find x.</td><td>1. 38</td></tr>
<tr><td>2. 30</td></tr>
<tr><td>3. 28</td></tr>
<tr><td>4. 17</td></tr>
<tr><td>5. 27</td></tr>
<tr><td rowspan="5">Assignment 4
Find the sum-of-products expansion of the Boolean function $f(x, y)$ that has the value 1 if and only if $x = 1$, $y = 0$.</td><td>1. $x \wedge \bar{y}$</td></tr>
<tr><td>2. $\bar{x} \wedge y$</td></tr>
<tr><td>3. $x \vee \bar{y}$</td></tr>
<tr><td>4. $\bar{x} \vee y$</td></tr>
<tr><td>5. $x \wedge \bar{y}$</td></tr>
<tr><td rowspan="5">Assignment 5
Simplify the sum-of-products expansion of the Boolean function represented by the following K-map:
<table><tr><td></td><td>y</td><td>y</td><td>$\bar{y}$</td><td>$\bar{y}$</td><td></td></tr><tr><td>x</td><td>1</td><td>1</td><td></td><td></td><td>w</td></tr><tr><td>x</td><td>1</td><td>1</td><td></td><td></td><td>$\bar{w}$</td></tr><tr><td>$\bar{x}$</td><td>1</td><td>1</td><td></td><td></td><td>$\bar{w}$</td></tr><tr><td>$\bar{x}$</td><td>1</td><td>1</td><td></td><td></td><td>w</td></tr><tr><td></td><td>z</td><td>$\bar{z}$</td><td>$\bar{z}$</td><td>z</td><td></td></tr></table></td><td>1. $x \wedge y$</td></tr>
<tr><td>2. x</td></tr>
<tr><td>3. z</td></tr>
<tr><td>4. w</td></tr>
<tr><td>5. y</td></tr>
</table>

Assignment 6 Determine all edges of the following graph incident to vertex a:  	1. $\{a,d\},\{a,c\}$ 2. $\{c,b\},\{c,d\},\{e,d\}$ 3. $\{c,d\},\{a,c\}$ 4. $\{a,b\},\{a,c\}$ 5. Such edges don't exist.
Assignment 7 Determine the depth of the rooted tree in the figure:  	1. 6 2. 3 3. 4 4. 5 5. 9
Assignment 8 An encoding function is defined by the generator matrix $G = \begin{bmatrix} 1\,0\,0\,1\,1\,0 \\ 0\,1\,0\,1\,0\,1 \\ 0\,0\,1\,0\,0\,0 \end{bmatrix}$. Encode 100.	1. 011010 2. 100111 3. 011101 4. 100110 5. 111100
Assignment 9 Find a regular set corresponding to the regular expression $a(b \vee c)d$.	1. $\{aba, aca, ada\}$ 2. $\{abd, acd\}$ 3. $\{ac, ad, bc, bd\}$ 4. $\{acb, adb, bcb, bdb\}$ 5. $\{aac, aad, abc, abd\}$
Assignment 10 Determine which language is accepted by the automaton:	1. All words which do not have three consecutive bs. 2. All words which have three consecutive bs. 3. All words which do



not have three consecutive as.

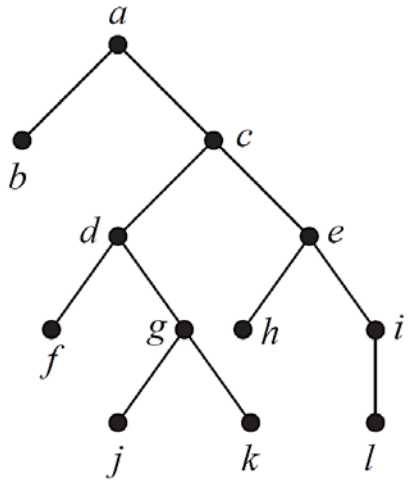
4. All words which have three consecutive as.

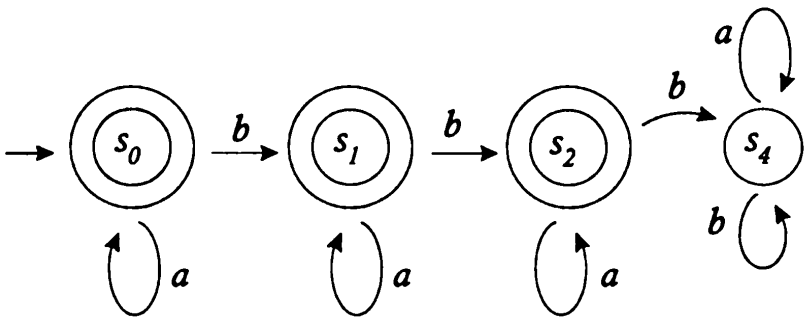
5. All words which have four consecutive bs.

Assignment 11

Set operations.

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the contrapositive of this conditional statement. If it snows today, I will sky tomorrow.</td><td>1. I will sky tomorrow only if it snows today.</td></tr>
<tr><td>2. If I do not sky tomorrow, then it will not have snowed today.</td></tr>
<tr><td>3. If it does not snow today, then I will not sky tomorrow.</td></tr>
<tr><td>4. If it snows today, then I will not sky tomorrow.</td></tr>
<tr><td>5. If it does not snow today, then I will sky tomorrow.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{a, b, c, d\}$ and $B = \{a, b, e, f, g, h\}$. Find $B \setminus A$.</td><td>1. $\{e, f, g, h\}$</td></tr>
<tr><td>2. $\{f, g, h\}$</td></tr>
<tr><td>3. $\{c, d\}$</td></tr>
<tr><td>4. $\varnothing$</td></tr>
<tr><td>5. $\{a, b, c, g, h\}$</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of three-digit numbers made from the digits 1, 2, 3, 6 and 8. How many numbers are there in S?</td><td>1. 125</td></tr>
<tr><td>2. 5</td></tr>
<tr><td>3. 50</td></tr>
<tr><td>4. 15</td></tr>
<tr><td>5. 300</td></tr>
<tr><td rowspan="5">Assignment 4
Find the product-of-sums expansion of the Boolean function $f(x, y)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>$f(x, y)$</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table></td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge y)$</td></tr>
<tr><td>2. $(\bar{x} \wedge y) \vee (x \wedge \bar{y})$</td></tr>
<tr><td>3. $(x \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>4. $(\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>5. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td rowspan="5">Assignment 5
Find the dual of the Boolean function $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 0)$.</td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 1)$</td></tr>
<tr><td>2. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 0)$</td></tr>
<tr><td>3. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 0)$</td></tr>
<tr><td>4. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 1)$</td></tr>
<tr><td>5. $(\bar{x} \vee \bar{y}) \vee (\bar{x} \vee 0)$</td></tr>
<tr><td rowspan="5">Assignment 6
Let A $= \{a, b\}$. Which of the following relations on a set A are reflexive?
$S = \{(a, a), (a, b), (b, a)\}$,
$T = \{(a, a), (a, b), (b, b)\}$,
$U = \{(a, b), (b, a)\}$,
$V = \{(b, b), (a, b), (b, a)\}$.</td><td>1. S, T, U, V</td></tr>
<tr><td>2. T</td></tr>
<tr><td>3. U</td></tr>
<tr><td>4. S, U</td></tr>
<tr><td>5. T, U</td></tr>
</table>

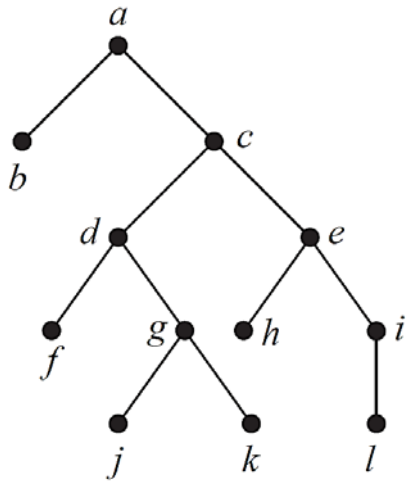
<table>
<tr>
<td>Assignment 7
Write down all leaves of the rooted tree in the figure:
 
</td>
<td>1. b, f, j, h, l
2. d, f, i
3. b, f, j, h, l, k
4. d, f, g
5. b, c, e, i</td>
</tr>
<tr>
<td>Assignment 8
Use the coding scheme
<table><tr><th>Символ</th><th>Код</th></tr><tr><td>a</td><td>111</td></tr><tr><td>c</td><td>101</td></tr><tr><td>e</td><td>01</td></tr><tr><td>i</td><td>11011</td></tr><tr><td>g</td><td>0010</td></tr><tr><td>h</td><td>1001</td></tr><tr><td>m</td><td>0001</td></tr><tr><td>s</td><td>10001</td></tr><tr><td>l</td><td>11001</td></tr></table>to encode the word $mammals$.</td>
<td>1. 11110110111100110000100110001
2. 1111101010110000110001
3. 10101011100101010101
4. 1010111001010001010
5. 000111100010001111110011000 1</td>
</tr>
<tr>
<td>Assignment 9
Find a regular expression corresponding to the regular set $\{ba, ca, da\}$.</td>
<td>1. $(a \vee b)(b \vee c)$
2. $a(\lambda \vee c \vee d)b$
3. $(a \vee b \vee c)(d \vee e \vee f)$
4. $(b \vee c \vee d)a$
5. $a(b \vee c \vee d)$</td>
</tr>
<tr>
<td>Assignment 10
Which of the following words are not accepted by the following automaton?</td>
<td>1. b
2. bb
3. bba
4. $bbba$
5. aa</td>
</tr>
</table>

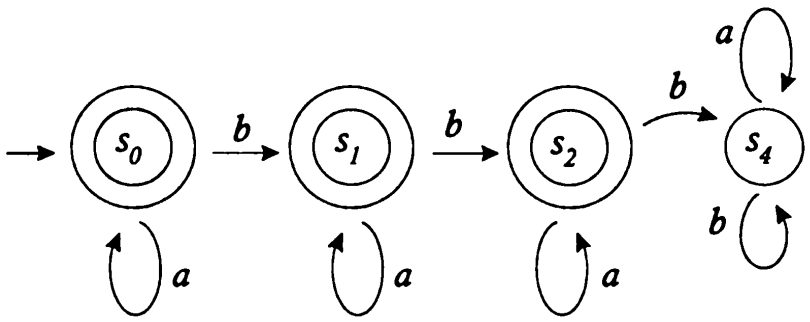


Assignment 11

Propositions. Compound propositions. Conditional statements.

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the inverse of this conditional statement. If it snows today, I will sky tomorrow.</td><td>1. I will sky tomorrow only if it snows today.</td></tr>
<tr><td>2. If I do not sky tomorrow, then it will not have snowed today.</td></tr>
<tr><td>3. If it does not snow today, then I will not sky tomorrow.</td></tr>
<tr><td>4. If it snows today, then I will not sky tomorrow.</td></tr>
<tr><td>5. If it does not snow today, then I will sky tomorrow.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find $B \setminus A$.</td><td>1.{1, 2, 4, 5}</td></tr>
<tr><td>2.{0, 6}</td></tr>
<tr><td>3.{0, 1, 2, 3, 4, 5, 6}</td></tr>
<tr><td>4.{3, 6}</td></tr>
<tr><td>5.{3}</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of four-digit numbers made from the digits 6, 7 and 9. How many numbers are there in S?</td><td>1. 27</td></tr>
<tr><td>2. 81</td></tr>
<tr><td>3. 3</td></tr>
<tr><td>4. 15</td></tr>
<tr><td>5. 9</td></tr>
<tr><td rowspan="5">Assignment 4
Find the product-of-sums expansion of the Boolean function $f(x, y)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>$f(x,y)$</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table></td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge y)$</td></tr>
<tr><td>2. $(\bar{x} \wedge y) \vee (x \wedge \bar{y})$</td></tr>
<tr><td>3. $(x \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>4. $(\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>5. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td rowspan="5">Assignment 5
Find the dual of the Boolean function $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 1)$.</td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 1)$</td></tr>
<tr><td>2. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 0)$</td></tr>
<tr><td>3. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 0)$</td></tr>
<tr><td>4. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 1)$</td></tr>
<tr><td>5. $(\bar{x} \vee \bar{y}) \vee (\bar{x} \vee 1)$</td></tr>
<tr><td rowspan="5">Assignment 6
The characteristic equation of a linear homogeneous recurrence relation with constant coefficients (for a sequence $\{a_n\}$) is of the form: $(r - 2) \cdot (r - 3)$.
What is the general solution of this recurrence relation?</td><td>1. $a_n = C \cdot 6^n$</td></tr>
<tr><td>2. $a_n = C_1 \cdot 2^n + C_2 \cdot 3^n$</td></tr>
<tr><td>3. $a_n = C \cdot 2^n$</td></tr>
<tr><td>4. $a_n = C \cdot 3^n$</td></tr>
<tr><td>5. $a_n = 10 \cdot (2^n + 3^n)$</td></tr>
</table>

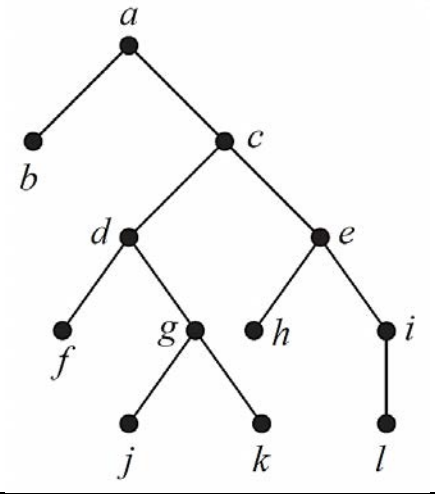
<table>
<tr>
<td>Assignment7
Write down all internal vertices of the rooted tree in the figure:
 
</td>
<td>1. d, f, g, h, j
2. b, c, i
3. a, b
4. d, f, g
5. c, d, e, i, g</td>
</tr>
<tr>
<td>Assignment 8
Use the coding scheme
<table><tr><th>Символ</th><th>Код</th></tr><tr><td>a</td><td>111</td></tr><tr><td>c</td><td>101</td></tr><tr><td>e</td><td>01</td></tr><tr><td>i</td><td>11011</td></tr><tr><td>g</td><td>0010</td></tr><tr><td>h</td><td>1001</td></tr><tr><td>m</td><td>0001</td></tr><tr><td>s</td><td>10001</td></tr><tr><td>t</td><td>11001</td></tr></table>to encode the word $agate$.</td>
<td>1. 111101101111001100010011
2. 11100101111100101
3. 1010101110010101010101
4. 1010111001010001010
5. 000111100010001111110011 0</td>
</tr>
<tr>
<td>Assignment 9
Find a regular expression corresponding to the regular set $\{ab, ac, ad, bb, bc, bd\}$.</td>
<td>1. $(a \vee b)(b \vee c)$
2. $a(\lambda \vee c \vee d)b$
3. $(a \vee b \vee c)(d \vee e \vee \mathrm{f})$
4. $(b \vee c \vee d)a$
5. $(a \vee b)(b \vee c \vee d)$</td>
</tr>
<tr>
<td>Assignment 10
Which of the following words are accepted by the following automaton?</td>
<td>1. $bbba$
2. aaa
3. $abbba$
4. $abaabab$
5. $abbbbaa$</td>
</tr>
</table>



Assignment 11

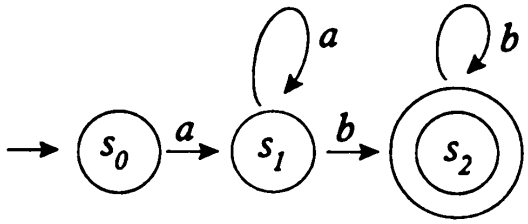
Propositional satisfiability. Satisfiability problem.

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the converse of this conditional statement. I come to class whenever there is going to be a quiz.</td><td>1. If I do not come to class, then there will be a quiz.</td></tr>
<tr><td>2. If I come to class, then there will be a quiz.</td></tr>
<tr><td>3. If I come to class, then there will not be a quiz.</td></tr>
<tr><td>4. If I do not come to class, then there will not be a quiz.</td></tr>
<tr><td>5. If there is not going to be a quiz, then I don’t come to class.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$ and $C = \{4, 5, 6, 7, 8, 9, 10\}$.
Find $A \cap B \cap C$.</td><td>1. {4, 6}</td></tr>
<tr><td>2. {0, 2, 4, 6}</td></tr>
<tr><td>3. {4, 5, 6}</td></tr>
<tr><td>4. {4, 5, 6, 7 , 8, 9, 10}</td></tr>
<tr><td>5. $\varnothing$</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of three-digit numbers made from the digits 6, 7, 8 and 9. How many numbers in S have no repeated digit?</td><td>1. 81</td></tr>
<tr><td>2. 4</td></tr>
<tr><td>3. 24</td></tr>
<tr><td>4. 1000</td></tr>
<tr><td>5. 27</td></tr>
<tr><td rowspan="5">Assignment 4
Find the sum-of-products expansion of the Boolean function $f(x, y, z)$, which is given in the table:
<table><tr><th>x</th><th>y</th><th>z</th><th>$f(x, y, z)$</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table></td><td>1. $x \wedge \bar{y} \wedge z$</td></tr>
<tr><td>2. $(\bar{x} \wedge y \wedge \bar{z}) \vee (x \wedge \bar{y} \wedge z)$</td></tr>
<tr><td>3. $\bar{x} \vee y \vee \bar{z}$</td></tr>
<tr><td>4. $x \vee \bar{y} \vee z$</td></tr>
<tr><td>5. $(x \wedge y \wedge z) \vee (x \wedge \bar{y} \wedge \bar{z})$</td></tr>
<tr><td rowspan="5">Assignment 5
Simplify the sum-of-products expansion of the Boolean function represented by the following K-map:
<table><tr><td></td><td>x</td><td>$\bar{x}$</td></tr><tr><td>y</td><td>1</td><td>1</td></tr><tr><td>$\bar{y}$</td><td>1</td><td></td></tr></table></td><td>1. $\bar{x} \vee \bar{y}$</td></tr>
<tr><td>2. y</td></tr>
<tr><td>3. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td>4. x</td></tr>
<tr><td>5. $x \vee y$</td></tr>
</table>

<table>
<tr><td>Assignment 6
Let A $= \{a, b\}$. Which of the following relations on a set A are symmetric?
$S = \{(a,a),(a,b),(b,a)\}$,
$T = \{(a,a),(a,b),(b,b)\}$,
$U = \{(a,b),(b,a)\}$,
$V = \{(b,b),(a,b),(b,a)\}$.</td><td>1. S, T, U, V
2. S, U, V
3. T
4. S, T
5. T, U</td></tr>
<tr><td>Assignment 7
Write down the left child of vertex c in the binary tree in the figure:
 
</td><td>1. a
2. d
3. e
4. f
5. g</td></tr>
<tr><td>Assignment 8
Use the coding scheme
<table><tr><th>Символ</th><th>Код</th></tr><tr><td>a</td><td>111</td></tr><tr><td>c</td><td>101</td></tr><tr><td>e</td><td>01</td></tr><tr><td>i</td><td>11011</td></tr><tr><td>g</td><td>0010</td></tr><tr><td>h</td><td>1001</td></tr><tr><td>m</td><td>0001</td></tr><tr><td>s</td><td>10001</td></tr><tr><td>t</td><td>11001</td></tr></table>to decode the word 11001100101000101.</td><td>1. $mathematics$
2. $gates$
3. $seismic$
4. $theme$
5. $agate$</td></tr>
<tr><td>Assignment 9
Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two bs.</td><td>1. $a^*ba^*ba^*$
2. $b^*ab^*ab^*$
3. $(a \cup c)^*b(a \cup c)^*b(a \cup c)^*$
4. $(b \cup c)^*a(b \cup c)^*a$
5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a$</td></tr>
</table>

Assignment 10

Which of the following words are not accepted by the following automaton?

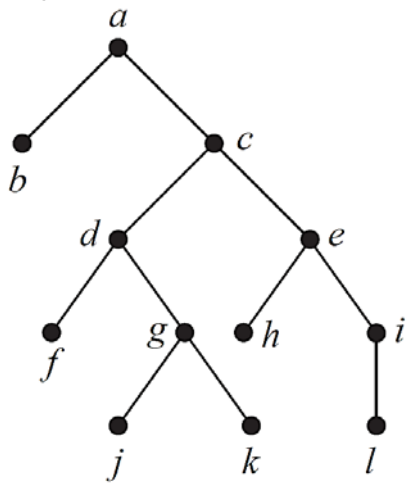


1. aab
2. $aabb$
3. abb
4. $aabbb$
5. aa

Assignment 11

Logical equivalences involving quantifiers.

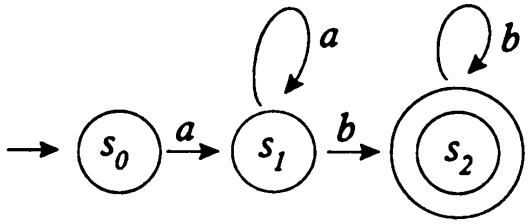
<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the contrapositive of this conditional statement. I come to class whenever there is going to be a quiz.</td><td>1. If I do not come to class, then there will be a quiz.</td></tr>
<tr><td>2. If I come to class, then there will be a quiz.</td></tr>
<tr><td>3. If I come to class, then there will not be a quiz.</td></tr>
<tr><td>4. If I do not come to class, then there will not be a quiz.</td></tr>
<tr><td>5. If there is not going to be a quiz, then I don’t come to class.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{0, 2, 4\}, B = \{0, 1, 2, 3\}$ and $C = \{4, 5\}$. Find $A \cup B \cup C$.</td><td>1. {0, 1, 2, 3, 4, 5}</td></tr>
<tr><td>2. {0, 2, 4, 5}</td></tr>
<tr><td>3. {0, 1, 2, 3, 4}</td></tr>
<tr><td>4. {0, 1, 2, 3, 5}</td></tr>
<tr><td>5. $\varnothing$</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of three-digit numbers made from the digits 1, 6, 7, 8 and 9. How many numbers in S have no repeated digit?</td><td>1. 5</td></tr>
<tr><td>2. 125</td></tr>
<tr><td>3. 1000</td></tr>
<tr><td>4. 75</td></tr>
<tr><td>5. 60</td></tr>
<tr><td rowspan="5">Assignment 4
Find the sum-of-products expansion of the Boolean function $f(x, y, z)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>z</th><th>$f(x, y, z)$</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table></td><td>1. $x \wedge y \wedge z$</td></tr>
<tr><td>2. $(\bar{x} \wedge y \wedge \bar{z}) \vee (x \wedge \bar{y} \wedge z)$</td></tr>
<tr><td>3. $\bar{x} \vee y \vee \bar{z}$</td></tr>
<tr><td>4. $x \vee \bar{y} \vee \bar{z}$</td></tr>
<tr><td>5. $(x \wedge y \wedge z) \vee (x \wedge \bar{y} \wedge \bar{z})$</td></tr>
<tr><td rowspan="5">Assignment 5
Simplify the sum-of-products expansion of the Boolean function represented by the following K-map:
<table><tr><th></th><th>x</th><th>$\bar{x}$</th></tr><tr><td>y</td><td>1</td><td>1</td></tr><tr><td>$\bar{y}$</td><td></td><td>1</td></tr></table></td><td>1. $\bar{x} \vee y$</td></tr>
<tr><td>2. y</td></tr>
<tr><td>3. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td>4. x</td></tr>
<tr><td>5. $x \vee y$</td></tr>
</table>

Assignment 6 The characteristic equation of a linear homogeneous recurrence relation with constant coefficients (for a sequence $\{a_n\}$) is of the form: $(r-5)\cdot(r-2)$. What is the general solution of this recurrence relation?	1. $a_n = C_1 \cdot 2^n + C_2 \cdot 5^n$ 2. $a_n = C \cdot 10^n$ 3. $a_n = C \cdot 2^n$ 4. $a_n = C \cdot 5^n$ 5. $a_n = 10 \cdot (2^n + 5^n)$
Assignment 7 Write down the right child of vertex c in the following binary tree:  	1. a 2. d 3. e 4. f 5. g
Assignment 8 Use the coding scheme to decode the word 11100101111100101.	1. $mathematics$ 2. $gates$ 3. $seismic$ 4. $theme$ 5. $agate$
Assignment 9 Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two as.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b$ 4. $(b \cup c)^*a(b \cup c)^*a(b \cup c)^*$ 5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a(a \cup b \cup c)^*$

Символ	Код
a	111
c	101
e	01
i	11011
g	0010
h	1001
m	0001
s	10001
t	11001

Assignment 10

Which of the following words are accepted by the following automaton?

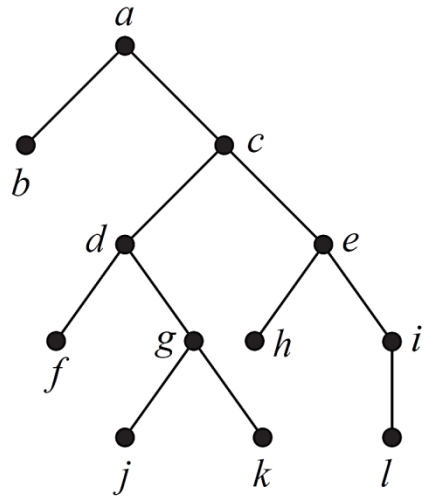
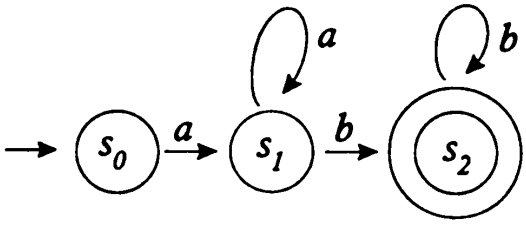


1. aa
2. aaa
3. abb
4. $aaaa$
5. ba

Assignment 11

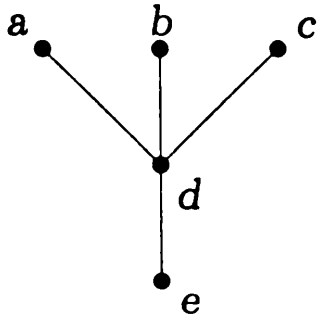
Negating quantified expressions.

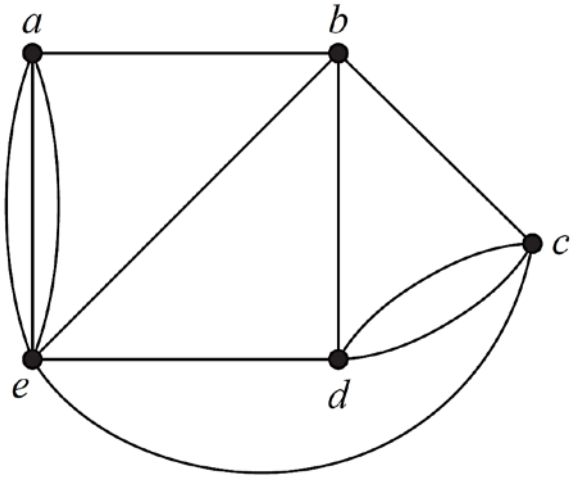
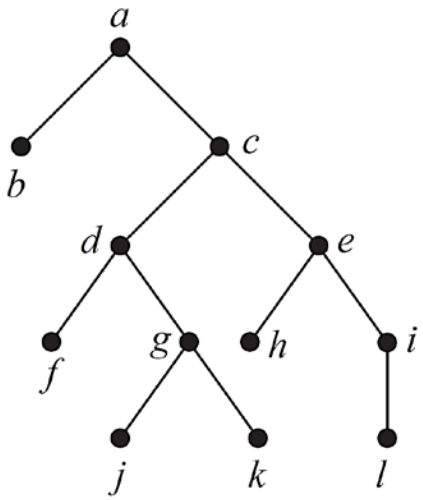
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Suppose that the domain of the propositional function $P(x)$ consists of the integers 1, 2 and 3. Write out the proposition $\exists x P(x)$ using conjunctions ($\wedge$), disjunctions ($\vee$) and negations ($^{-}$).</td><td>1. $P(1) \vee P(2) \vee P(3)$</td></tr>
<tr><td></td><td>2. $(P(1) \wedge P(2)) \vee P(3)$</td></tr>
<tr><td></td><td>3. $P(1) \wedge P(2) \wedge P(3)$</td></tr>
<tr><td></td><td>4. $\overline{P(1)} \wedge \overline{P(2)} \wedge \overline{P(3)}$</td></tr>
<tr><td></td><td>5. $\overline{P(1)} \vee \overline{P(2)} \vee \overline{P(3)}$</td></tr>
<tr><td>Assignment 2
Let $A = \{a\}$ and $B = \{y, z\}$. Find $A \times B$.</td><td>1. $(a, y), (a, z)$</td></tr>
<tr><td></td><td>2. a, y, z</td></tr>
<tr><td></td><td>3. $(a, a), (y, y), (z, z)$</td></tr>
<tr><td></td><td>4. $(a, y), z$</td></tr>
<tr><td></td><td>5. $(a, a), (y, z), (z, y)$</td></tr>
<tr><td>Assignment 3
Find the coefficient of $a^3 b^5$ in the expansion $(a + b)^8$.</td><td>1. 78</td></tr>
<tr><td></td><td>2. 56</td></tr>
<tr><td></td><td>3. 8</td></tr>
<tr><td></td><td>4. 1</td></tr>
<tr><td></td><td>5. 4</td></tr>
<tr><td>Assignment 4
Find the product-of-sums expansion of the Boolean function $f(x, y, z)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>z</th><th>$f(x, y, z)$</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table></td><td>1. $x \vee \bar{y} \vee z$</td></tr>
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<tr><td></td><td>3. $\bar{x} \vee y \vee \bar{z}$</td></tr>
<tr><td></td><td>4. $(x \vee \bar{y} \vee z) \wedge (\bar{x} \vee y \vee \bar{z})$</td></tr>
<tr><td></td><td>5. $(\bar{x} \vee y \vee z) \wedge (\bar{x} \vee \bar{y} \vee \bar{z})$</td></tr>
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<tr><td></td><td>2. y</td></tr>
<tr><td></td><td>3. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td></td><td>4. x</td></tr>
<tr><td></td><td>5. $x \vee y$</td></tr>
<tr><td>Assignment 6
Let A $= \{a, b\}$. Which of the following</td><td>1. S, T, U, V</td></tr>
<tr><td></td><td>2. S, U, V</td></tr>
</table>

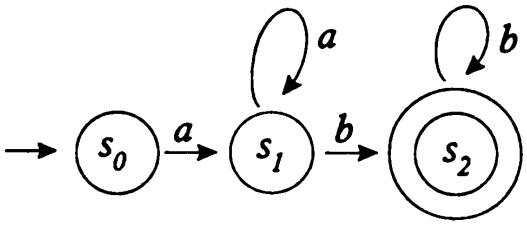
relations on a set A are reflexive and transitive? $S = \{(a,a),(a,b),(b,a)\}$, $T = \{(a,a),(a,b),(b,b)\}$, $U = \{(a,b),(b,a)\}$, $V = \{(b,b),(a,b),(b,a)\}$.	3. T 4. S 5. T, U
Assignment 7 Determine the depth of the rooted tree in the figure:  	1. 2 2. 3 3. 4 4. 5 5. 11
Assignment 8 An encoding function is defined by the generator matrix $G = \begin{bmatrix} 1\,0\,0\,1\,1\,1 \\ 0\,1\,0\,1\,0\,1 \\ 0\,0\,1\,1\,1\,1 \end{bmatrix}$. Encode 011.	1. 011010 2. 100111 3. 011101 4. 100110 5. 111100
Assignment 9 Let $A = \{a,b,c\}$. Give a regular expression for the set of all elements of A^*, containing exactly two bs.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b(a \cup c)^*$ 4. $(b \cup c)^*a(b \cup c)^*a(b \cup c)^*$ 5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a(a \cup b \cup c)^*$
Assignment 10 Find a regular expression for the language accepted by the automaton:  	1. $a^*ba^*ba^*$ 2. $ab(aab)^*$ 3. aa^*bb^* 4. a^*ba^* 5. $a^*ba^*ba^*b$

Assignment 11

Computer representation of sets.

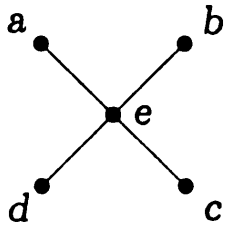
Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x, y)$ consists of pairs (x, y), where $x = 1, 2$ and $y = 0, 2, 3$. Write out the proposition $\forall x P(x, 3)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(1,3) \vee P(2,3)$ 2. $P(1,3) \wedge P(2,3)$ 3. $P(2,2) \vee P(2,3)$ 4. $P(2,0) \vee P(2,3)$ 5. $P(2,0) \wedge P(2,2)$
Assignment 2 List all elements incomparable with d in the partial ordering with the accompanying Hasse diagram.  	1. a, b, c 2. e 3. d 4. a, b, c, e 5. Such elements don't exist.
Assignment 3 A sequence of integers is defined by $f(0) = 1$, $f(1) = 3, f(k) = 2 \cdot f(k-1) - f(k-2)$. Calculate $f(3)$.	1. 1 2. 3 3. 7 4. 16 5. 5
Assignment 4 The Boolean operator $\oplus$, called the XOR operator, is defined by $x \oplus y = (\bar{x} \wedge y) \vee (x \wedge \bar{y})$. Simplify the expression $x \oplus x$.	1. 0 2. 1 3. x 4. $\bar{x}$ 5. $x \oplus x$
Assignment 5 Simplify the Boolean function $f(x, y, z, w) = (x \wedge y \wedge z \wedge w) \vee (\bar{x} \wedge y \wedge z \wedge w)$.	1. $y \wedge z \wedge w$ 2. $x \wedge z \wedge w$ 3. $x \wedge y \wedge w$ 4. $x \wedge y \wedge z$ 5. $\bar{x} \wedge y \wedge z \wedge w$
Assignment 6 Find an Eulerian path in the Eulerian graph in the figure:	1. $edbae$ 2. $edbe$ 3. $edcdbe$ 4. $edbaeae$ 5. $edbaeaebcdce$

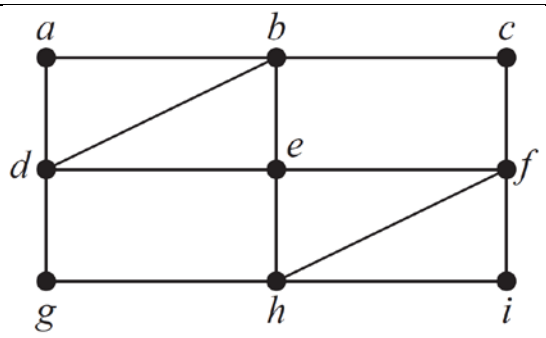
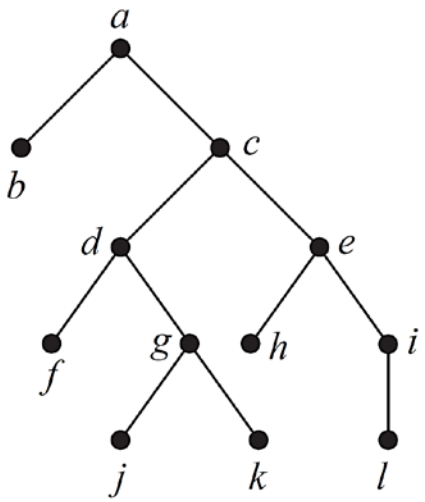
	
Assignment 7 Write down all leaves in the following binary rooted tree: 	1. e, g, h, i, j, k 2. b, c 3. b, f, j, k, h, l 4. d, e, f, g 5. e, g, h, i, j, k
Assignment 8 The minimum distance of the code is 3. Find the maximum number of errors which the code can detect.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Find a regular expression corresponding to the regular set $\{ba, ca, da\}$.	1. $(a \vee b)(b \vee c)$ 2. $a(\lambda \vee c \vee d)b$ 3. $(a \vee b \vee c)(d \vee e \vee f)$ 4. $(b \vee c \vee d)a$ 5. $a(b \vee c \vee d)$
Assignment 10 Find a regular expression for the language accepted by the automaton:	1. $a^*ba^*ba^*$ 2. $ab(aab)^*$ 3. aa^*bb^* 4. a^*ba^* 5. $a^*ba^*ba^*b$

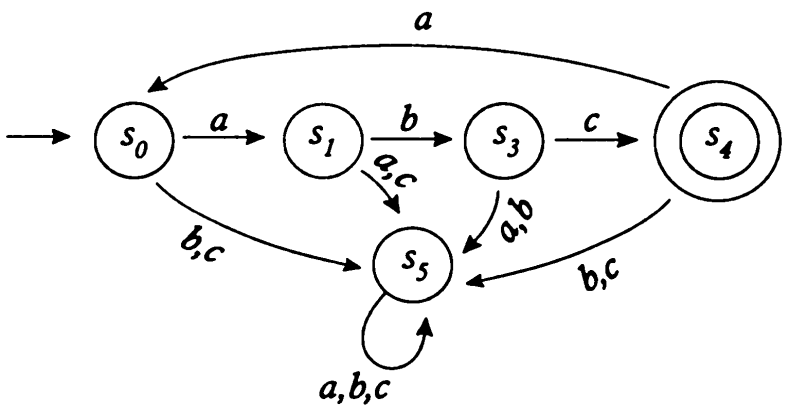


Assignment 11

The pigeonhole principle.

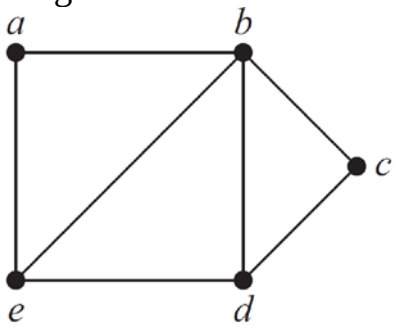
Test assignments	Answers
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Assignment 2 List all elements incomparable with a in the partial ordering with the accompanying Hasse diagram.  	1. e, b, c 2. b 3. d, c 4. c, d, b 5. Such elements don't exist.
Assignment 3 A sequence of integers is defined by $f(0) = 2$, $f(1) = 4, f(k) = 3 \cdot f(k-1) - 2 \cdot f(k-2)$. Calculate $f(3)$.	1. 24 2. 8 3. 16 4. 4 5. 6
Assignment 4 The Boolean operator $\oplus$, called the XOR operator, is defined by $x \oplus y = (\bar{x} \wedge y) \vee (x \wedge \bar{y})$. Simplify the expression $x \oplus \bar{x}$.	1. 0 2. 1 3. x 4. $\bar{x}$ 5. $x \oplus \bar{x}$
Assignment 5 Simplify the Boolean function $f(x, y, z, w) = (x \wedge y \wedge z \wedge w) \vee (x \wedge \bar{y} \wedge z \wedge w)$.	1. $y \wedge z \wedge w$ 2. $x \wedge z \wedge w$ 3. $x \wedge y \wedge w$ 4. $x \wedge y \wedge z$ 5. $x \wedge \bar{y} \wedge z \wedge w$
Assignment 6 Find an Eulerian path in the Eulerian graph in the figure:	1. $fihefhef$ 2. $fedbehfcbadghif$ 3. $fcbadghif$ 4. $fedbehf$ 5. $fcbef$

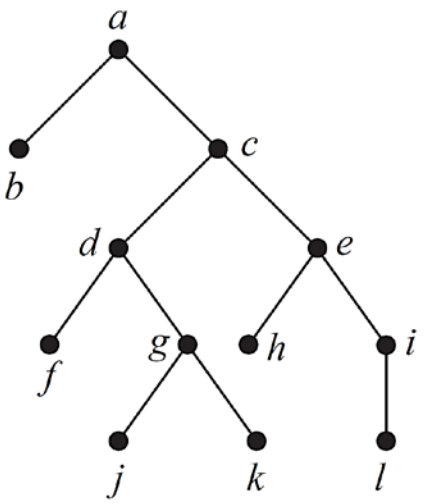
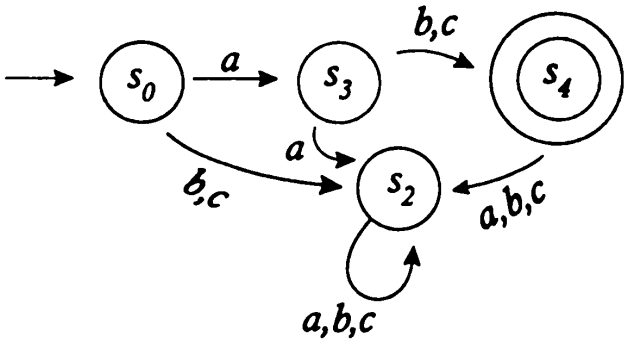
	
Assignment 7 Write down all internal vertices of the following rooted tree 	1. d, f, g, h, j, k 2. b, c, e, i, d, g 3. c, e, i, d, g 4. d, f, g, j, k 5. b, c, d, f
Assignment 8 The minimum distance of the code is 3. Find the maximum number of errors which the code can correct.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Find a regular expression corresponding to the regular set $\{ab, ac, ad, bb, bc, bd\}$.	1. $(a \vee b)(b \vee c)$ 2. $a(\lambda \vee c \vee d)b$ 3. $(a \vee b \vee c)(d \vee e \vee f)$ 4. $(b \vee c \vee d)a$ 5. $(a \vee b)(b \vee c \vee d)$
Assignment 10 Find a regular expression for the language accepted by the automaton:	1. $a^*ba^*ba^*$ 2. $abc(aabc)^*$ 3. $a(b \vee c)$ 4. a^*ba^*c 5. $a^*ba^*ba^*c$

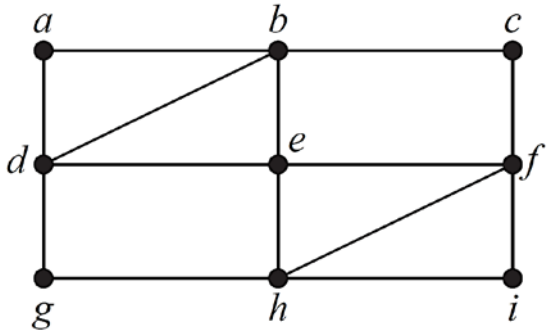


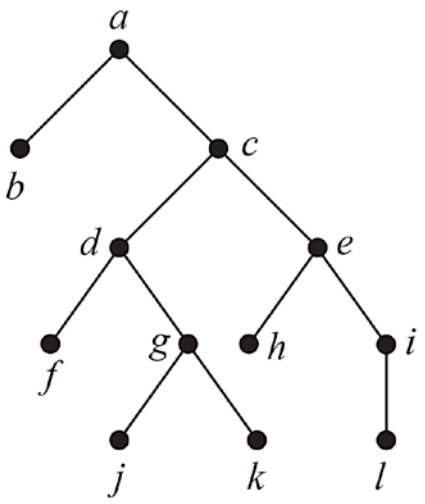
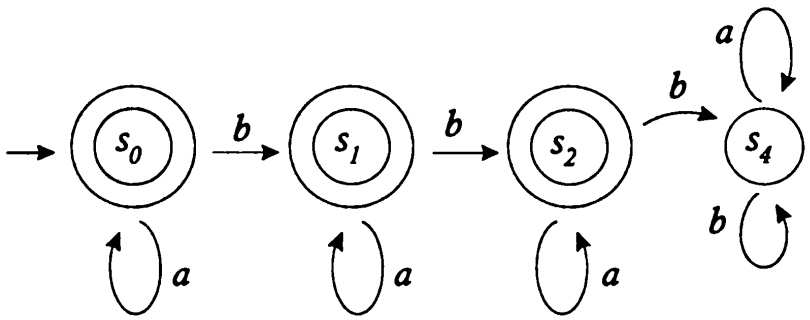
Assignment 11

The principle of inclusion-exclusion.

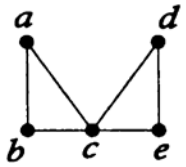
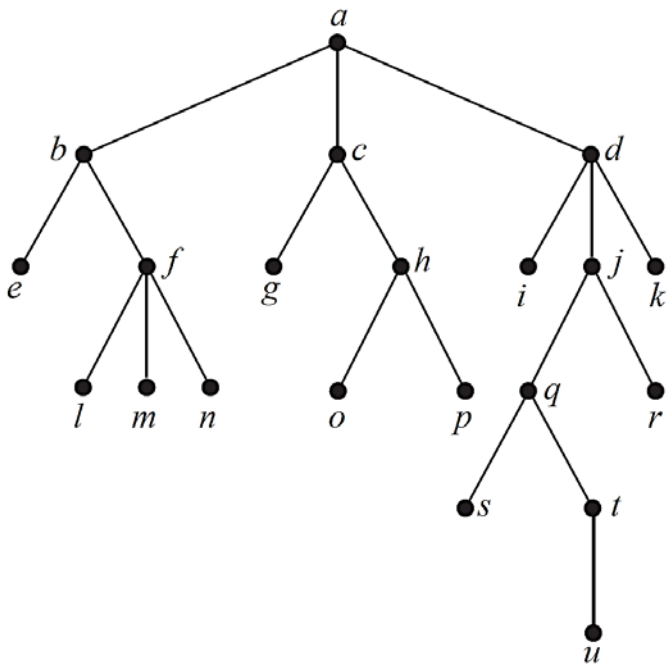
Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x,y)$ consists of pairs (x,y), where $x=1,2$ and $y=0,2,3$. Write out the proposition $\forall y P(2,y)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(2,0)\vee P(2,2)\vee P(2,3)$ 2. $P(2,0)\wedge P(2,2)\wedge P(2,3)$ 3. $P(2,2)\vee P(2,3)$ 4. $P(2,0)\vee P(2,3)$ 5. $P(2,0)\wedge P(2,2)$
Assignment 2 Suppose that $A=\{1,2,3,4,5,6,7,8,9,10\}$. Let R be the relation on the set A such that $(x,y)\in R$ if and only if $x-y$ is an even integer. R is an equivalence relation. What is the equivalence class [3] for this equivalence relation?	1. $\{2,4,6,8,10\}$ 2. $\{1,3,5,7,9\}$ 3. $\{6\}$ 4. $\{3,6,9\}$ 5. $\emptyset$
Assignment 3 Find the number of elements in the union $A_1\cup A_2\cup A_3$ if there are 100 elements in each set A_1,A_2,A_3 and these sets are pairwise disjoint.	1. 100 2. 300 3. 225 4. 200 5. 150
Assignment 4 The Boolean operator | , called the NAND operator, is defined by $x\mid y=\overline{x\wedge y}$. Express $\bar{x}$ using just the | operator.	1. $x\mid(x\mid x)$ 2. $\bar{x}\mid x$ 3. $x\mid x$ 4. $x\mid(\bar{x}\mid\bar{x})$ 5. $\bar{x}\mid(x\mid x)$
Assignment 5 Simplify the Boolean function $f(x,y,z,w)=(x\wedge y\wedge z\wedge w)\vee(x\wedge y\wedge\bar{z}\wedge w)$.	1. $y\wedge z\wedge w$ 2. $x\wedge z\wedge w$ 3. $x\wedge y\wedge w$ 4. $x\wedge y\wedge z$ 5. $x\wedge y\wedge\bar{z}\wedge w$
Assignment 6 Find a Hamiltonian cycle in the Hamiltonian graph in the figure:  	1. $ebcde$ 2. $ebcdbe$ 3. $abcdea$ 4. $ebde$ 5. $eabde$

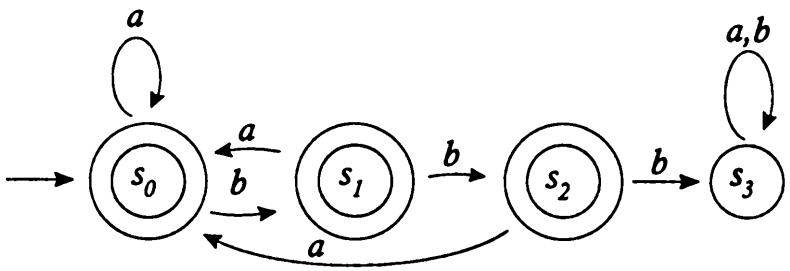
Assignment 7 Write down the left child of vertex d in the following binary tree:  	1. b 2. c 3. e 4. f 5. g
Assignment 8 The minimum distance of the code is 4. Find the maximum number of errors which the code can detect.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two bs.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b(a \cup c)^*$ 4. $(b \cup c)^*a(b \cup c)^*a$ 5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a$
Assignment 10 Find a regular expression for the language accepted by the automaton:  	1. $a^*ba^*ba^*$ 2. $abc(aabc)^*$ 3. $a(b \vee c)$ 4. a^*ba^*c 5. $a^*ba^*ba^*c$
Assignment 11 Equivalence relations.	

Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x,y)$ consists of pairs (x,y), where $x=1,2$ and $y=0,2,3$. Write out the proposition $\exists y P(2,y)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(2,0)\vee P(2,2)\vee P(2,3)$ 2. $P(2,0)\wedge P(2,2)\wedge P(2,3)$ 3. $P(2,2)\vee P(2,3)$ 4. $P(2,0)\vee P(2,3)$ 5. $P(2,0)\wedge P(2,2)$
Assignment 2 Suppose that $A=\{1,2,3,4,5,6,7,8,9,10\}$. Let R be the relation on the set A such that $(x,y)\in R$ if and only if $x-y$ is an even integer. R is an equivalence relation. What is the equivalence class $[6]$ for this equivalence relation?	1. $\{2,4,6,8,10\}$ 2. $\{1,3,5,7,9\}$ 3. $\{6\}$ 4. $\{3,6,9\}$ 5. $\emptyset$
Assignment 3 There are x computer science students at a university. Of these, 94 have taken a course in $C^{\#}$, 70 have taken a course in C^{++} and 81 have taken a course in $Java$. Each computer science student has taken a course only in one programming language. Find x.	1. 151 2. 175 3. 245 4. 164 5. 140
Assignment 4 The Boolean operator $\downarrow$, called the NOR operator, is defined by $x\downarrow y=\overline{x\vee y}$. Express $\bar{x}$ using just the $\downarrow$ operator.	1. $x\downarrow(x\downarrow x)$ 2. $\bar{x}\downarrow x$ 3. $x\downarrow x$ 4. $x\downarrow(\bar{x}\downarrow\bar{x})$ 5. $\bar{x}\downarrow(x\downarrow x)$
Assignment 5 Simplify the Boolean function $f(x,y,z,w)=(x\wedge y\wedge z\wedge w)\vee(x\wedge y\wedge z\wedge\bar{w})$.	1. $y\wedge z\wedge w$ 2. $x\wedge z\wedge w$ 3. $x\wedge y\wedge w$ 4. $x\wedge y\wedge z$ 5. $x\wedge y\wedge z\wedge\bar{w}$
Assignment 6 Find a Hamiltonian cycle in the Hamiltonian graph in the figure: 	1. $ecbe$ 2. $ebadghifce$ 3. $ecbae$ 4. $ecbaeighe$ 5. $efcbade$

Assignment 7 Write down the right child of vertex d in the following binary tree:  	1. a 2. c 3. e 4. f 5. g
Assignment 8 The minimum distance of the code is 5. Find the maximum number of errors which the code can correct.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two as.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b$ 4. $(b \cup c)^*a(b \cup c)^*a(b \cup c)^*$ 5. $(a \cup b \cup c)^*aa(a \cup b \cup c)^*$
Assignment 10 Find a regular expression for the language accepted by the automaton:  	1. $a^*ba^*ba^*$ 2. $abc(aabc)^*$ 3. $a(b \vee c)$ 4. a^*ba^*c 5. $a^*ba^*ba^*c$
Assignment 11 Partial orderings.	

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the converse of this conditional statement. If it snows today, I will sky tomorrow.</td><td>1. I will sky tomorrow only if it snows today.</td></tr>
<tr><td>2. If I do not sky tomorrow, then it will not have snowed today.</td></tr>
<tr><td>3. If it does not snow today, then I will not sky tomorrow.</td></tr>
<tr><td>4. If it snows today, then I will not sky tomorrow.</td></tr>
<tr><td>5. If it does not snow today, then I will sky tomorrow.</td></tr>
<tr><td rowspan="5">Assignment 2
A palindrome is a string of digits that reads the same backwards as forwards. How many different palindromes are there with six digits?</td><td>1. 900</td></tr>
<tr><td>2. 1000</td></tr>
<tr><td>3. 9000</td></tr>
<tr><td>4. 850</td></tr>
<tr><td>5. 10000</td></tr>
<tr><td rowspan="5">Assignment 3
Each of the x first-year students studying computing at Yanka Kupala Grodno State University can study a number of optional units. Of these, 25 chose to study the accounting option, 13 chose the business option and 8 studied both of these options/ Find x.</td><td>1. 38</td></tr>
<tr><td>2. 30</td></tr>
<tr><td>3. 28</td></tr>
<tr><td>4. 17</td></tr>
<tr><td>5. 27</td></tr>
<tr><td rowspan="5">Assignment 4
Find the sum-of-products expansion of the Boolean function $f(x, y)$ that has the value 1 if and only if $x = 1$, $y = 0$.</td><td>1. $x \wedge \bar{y}$</td></tr>
<tr><td>2. $\bar{x} \wedge y$</td></tr>
<tr><td>3. $x \vee \bar{y}$</td></tr>
<tr><td>4. $\bar{x} \vee y$</td></tr>
<tr><td>5. $x \wedge \bar{y}$</td></tr>
<tr><td rowspan="5">Assignment 5
Simplify the sum-of-products expansion of the Boolean function represented by the following K-map:
<table><tr><td></td><td>y</td><td>y</td><td>$\bar{y}$</td><td>$\bar{y}$</td><td></td></tr><tr><td>x</td><td>1</td><td>1</td><td></td><td></td><td>w</td></tr><tr><td>x</td><td>1</td><td>1</td><td></td><td></td><td>$\bar{w}$</td></tr><tr><td>$\bar{x}$</td><td>1</td><td>1</td><td></td><td></td><td>$\bar{w}$</td></tr><tr><td>$\bar{x}$</td><td>1</td><td>1</td><td></td><td></td><td>w</td></tr><tr><td></td><td>z</td><td>$\bar{z}$</td><td>$\bar{z}$</td><td>z</td><td></td></tr></table></td><td>1. $x \wedge y$</td></tr>
<tr><td>2. x</td></tr>
<tr><td>3. z</td></tr>
<tr><td>4. w</td></tr>
<tr><td>5. y</td></tr>
</table>

Assignment 6 Determine all edges of the following graph incident to vertex a:  	1. $\{a,d\},\{a,c\}$ 2. $\{c,b\},\{c,d\},\{e,d\}$ 3. $\{c,d\},\{a,c\}$ 4. $\{a,b\},\{a,c\}$ 5. Such edges don't exist.
Assignment 7 Determine the depth of the rooted tree in the figure:  	1. 6 2. 3 3. 4 4. 5 5. 9
Assignment 8 An encoding function is defined by the generator matrix $G = \begin{bmatrix} 1\,0\,0\,1\,1\,0 \\ 0\,1\,0\,1\,0\,1 \\ 0\,0\,1\,0\,0\,0 \end{bmatrix}$. Encode 100.	1. 011010 2. 100111 3. 011101 4. 100110 5. 111100
Assignment 9 Find a regular set corresponding to the regular expression $a(b \vee c)d$.	1. $\{aba, aca, ada\}$ 2. $\{abd, acd\}$ 3. $\{ac, ad, bc, bd\}$ 4. $\{acb, adb, bcb, bdb\}$ 5. $\{aac, aad, abc, abd\}$
Assignment 10 Determine which language is accepted by the automaton:	1. All words which do not have three consecutive bs. 2. All words which have three consecutive bs. 3. All words which do



not have three consecutive as.

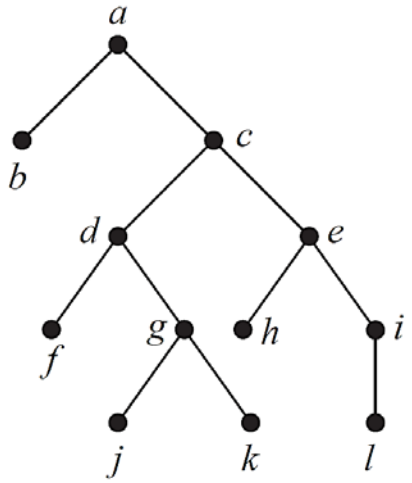
4. All words which have three consecutive as.

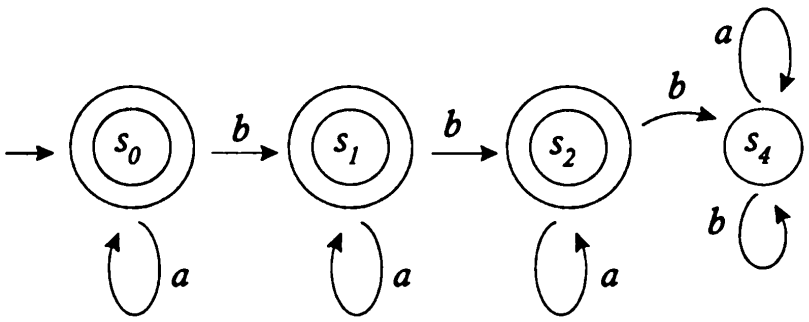
5. All words which have four consecutive bs.

Assignment 11

The topological sorting algorithm.

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the contrapositive of this conditional statement. If it snows today, I will sky tomorrow.</td><td>1. I will sky tomorrow only if it snows today.</td></tr>
<tr><td>2. If I do not sky tomorrow, then it will not have snowed today.</td></tr>
<tr><td>3. If it does not snow today, then I will not sky tomorrow.</td></tr>
<tr><td>4. If it snows today, then I will not sky tomorrow.</td></tr>
<tr><td>5. If it does not snow today, then I will sky tomorrow.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{a, b, c, d\}$ and $B = \{a, b, e, f, g, h\}$. Find $B \setminus A$.</td><td>1. $\{e, f, g, h\}$</td></tr>
<tr><td>2. $\{f, g, h\}$</td></tr>
<tr><td>3. $\{c, d\}$</td></tr>
<tr><td>4. $\varnothing$</td></tr>
<tr><td>5. $\{a, b, c, g, h\}$</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of three-digit numbers made from the digits 1, 2, 3, 6 and 8. How many numbers are there in S?</td><td>1. 125</td></tr>
<tr><td>2. 5</td></tr>
<tr><td>3. 50</td></tr>
<tr><td>4. 15</td></tr>
<tr><td>5. 300</td></tr>
<tr><td rowspan="5">Assignment 4
Find the product-of-sums expansion of the Boolean function $f(x, y)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>$f(x, y)$</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table></td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge y)$</td></tr>
<tr><td>2. $(\bar{x} \wedge y) \vee (x \wedge \bar{y})$</td></tr>
<tr><td>3. $(x \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>4. $(\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>5. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td rowspan="5">Assignment 5
Find the dual of the Boolean function $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 0)$.</td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 1)$</td></tr>
<tr><td>2. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 0)$</td></tr>
<tr><td>3. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 0)$</td></tr>
<tr><td>4. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 1)$</td></tr>
<tr><td>5. $(\bar{x} \vee \bar{y}) \vee (\bar{x} \vee 0)$</td></tr>
<tr><td rowspan="5">Assignment 6
Let A $= \{a, b\}$. Which of the following relations on a set A are reflexive?
$S = \{(a, a), (a, b), (b, a)\}$,
$T = \{(a, a), (a, b), (b, b)\}$,
$U = \{(a, b), (b, a)\}$,
$V = \{(b, b), (a, b), (b, a)\}$.</td><td>1. S, T, U, V</td></tr>
<tr><td>2. T</td></tr>
<tr><td>3. U</td></tr>
<tr><td>4. S, U</td></tr>
<tr><td>5. T, U</td></tr>
</table>

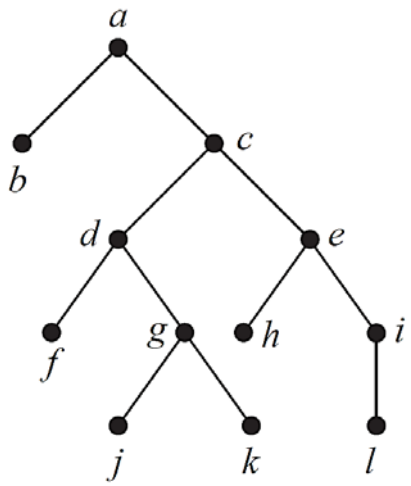
<table>
<tr>
<td>Assignment 7
Write down all leaves of the rooted tree in the figure:
 
</td>
<td>1. b, f, j, h, l
2. d, f, i
3. b, f, j, h, l, k
4. d, f, g
5. b, c, e, i</td>
</tr>
<tr>
<td>Assignment 8
Use the coding scheme
<table><tr><th>Символ</th><th>Код</th></tr><tr><td>a</td><td>111</td></tr><tr><td>c</td><td>101</td></tr><tr><td>e</td><td>01</td></tr><tr><td>i</td><td>11011</td></tr><tr><td>g</td><td>0010</td></tr><tr><td>h</td><td>1001</td></tr><tr><td>m</td><td>0001</td></tr><tr><td>s</td><td>10001</td></tr><tr><td>l</td><td>11001</td></tr></table>to encode the word $mammals$.</td>
<td>1. 11110110111100110000100110001
2. 1111101010110000110001
3. 10101011100101010101
4. 1010111001010001010
5. 000111100010001111110011000 1</td>
</tr>
<tr>
<td>Assignment 9
Find a regular expression corresponding to the regular set $\{ba, ca, da\}$.</td>
<td>1. $(a \vee b)(b \vee c)$
2. $a(\lambda \vee c \vee d)b$
3. $(a \vee b \vee c)(d \vee e \vee f)$
4. $(b \vee c \vee d)a$
5. $a(b \vee c \vee d)$</td>
</tr>
<tr>
<td>Assignment 10
Which of the following words are not accepted by the following automaton?</td>
<td>1. b
2. bb
3. bba
4. $bbba$
5. aa</td>
</tr>
</table>

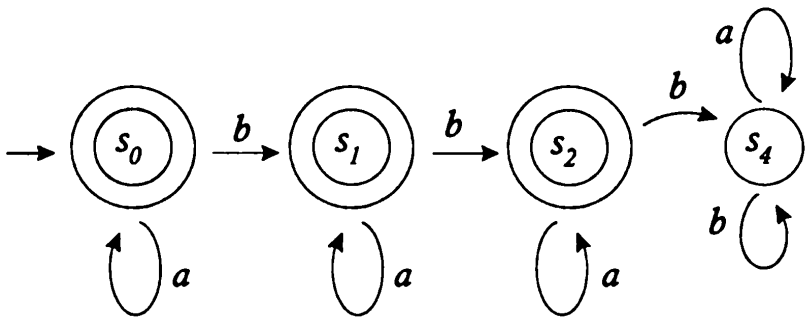


Assignment 11

The binomial theorem.

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the inverse of this conditional statement. If it snows today, I will sky tomorrow.</td><td>1. I will sky tomorrow only if it snows today.</td></tr>
<tr><td>2. If I do not sky tomorrow, then it will not have snowed today.</td></tr>
<tr><td>3. If it does not snow today, then I will not sky tomorrow.</td></tr>
<tr><td>4. If it snows today, then I will not sky tomorrow.</td></tr>
<tr><td>5. If it does not snow today, then I will sky tomorrow.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find $B \setminus A$.</td><td>1.{1, 2, 4, 5}</td></tr>
<tr><td>2.{0, 6}</td></tr>
<tr><td>3.{0, 1, 2, 3, 4, 5, 6}</td></tr>
<tr><td>4.{3, 6}</td></tr>
<tr><td>5.{3}</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of four-digit numbers made from the digits 6, 7 and 9. How many numbers are there in S?</td><td>1. 27</td></tr>
<tr><td>2. 81</td></tr>
<tr><td>3. 3</td></tr>
<tr><td>4. 15</td></tr>
<tr><td>5. 9</td></tr>
<tr><td rowspan="5">Assignment 4
Find the product-of-sums expansion of the Boolean function $f(x, y)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>$f(x,y)$</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table></td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge y)$</td></tr>
<tr><td>2. $(\bar{x} \wedge y) \vee (x \wedge \bar{y})$</td></tr>
<tr><td>3. $(x \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>4. $(\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>5. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td rowspan="5">Assignment 5
Find the dual of the Boolean function $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 1)$.</td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 1)$</td></tr>
<tr><td>2. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 0)$</td></tr>
<tr><td>3. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 0)$</td></tr>
<tr><td>4. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 1)$</td></tr>
<tr><td>5. $(\bar{x} \vee \bar{y}) \vee (\bar{x} \vee 1)$</td></tr>
<tr><td rowspan="5">Assignment 6
The characteristic equation of a linear homogeneous recurrence relation with constant coefficients (for a sequence $\{a_n\}$) is of the form: $(r - 2) \cdot (r - 3)$.
What is the general solution of this recurrence relation?</td><td>1. $a_n = C \cdot 6^n$</td></tr>
<tr><td>2. $a_n = C_1 \cdot 2^n + C_2 \cdot 3^n$</td></tr>
<tr><td>3. $a_n = C \cdot 2^n$</td></tr>
<tr><td>4. $a_n = C \cdot 3^n$</td></tr>
<tr><td>5. $a_n = 10 \cdot (2^n + 3^n)$</td></tr>
</table>

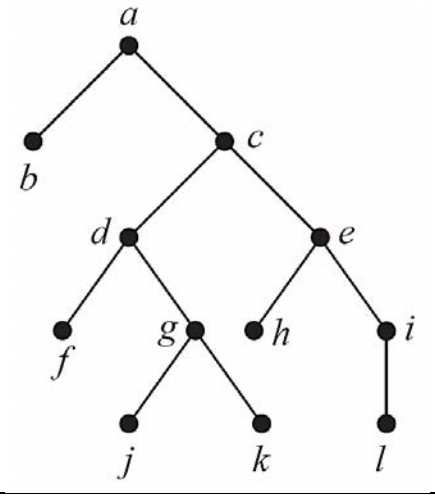
<table>
<tr>
<td>Assignment7
Write down all internal vertices of the rooted tree in the figure:
 
</td>
<td>1. d, f, g, h, j
2. b, c, i
3. a, b
4. d, f, g
5. c, d, e, i, g</td>
</tr>
<tr>
<td>Assignment 8
Use the coding scheme
<table><tr><th>Символ</th><th>Код</th></tr><tr><td>a</td><td>111</td></tr><tr><td>c</td><td>101</td></tr><tr><td>e</td><td>01</td></tr><tr><td>i</td><td>11011</td></tr><tr><td>g</td><td>0010</td></tr><tr><td>h</td><td>1001</td></tr><tr><td>m</td><td>0001</td></tr><tr><td>s</td><td>10001</td></tr><tr><td>t</td><td>11001</td></tr></table>to encode the word $agate$.</td>
<td>1. 111101101111001100010011
2. 11100101111100101
3. 1010101110010101010101
4. 1010111001010001010
5. 000111100010001111110011 0</td>
</tr>
<tr>
<td>Assignment 9
Find a regular expression corresponding to the regular set $\{ab, ac, ad, bb, bc, bd\}$.</td>
<td>1. $(a \vee b)(b \vee c)$
2. $a(\lambda \vee c \vee d)b$
3. $(a \vee b \vee c)(d \vee e \vee \mathrm{f})$
4. $(b \vee c \vee d)a$
5. $(a \vee b)(b \vee c \vee d)$</td>
</tr>
<tr>
<td>Assignment 10
Which of the following words are accepted by the following automaton?</td>
<td>1. $bbba$
2. aaa
3. $abbba$
4. $abaabab$
5. $abbbbaa$</td>
</tr>
</table>



Assignment 11

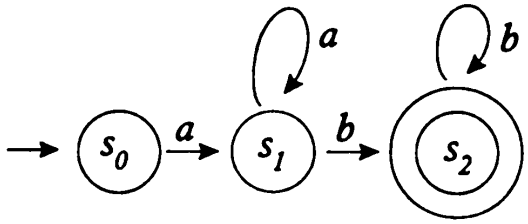
The topological sorting algorithm.

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the converse of this conditional statement. I come to class whenever there is going to be a quiz.</td><td>1. If I do not come to class, then there will be a quiz.</td></tr>
<tr><td>2. If I come to class, then there will be a quiz.</td></tr>
<tr><td>3. If I come to class, then there will not be a quiz.</td></tr>
<tr><td>4. If I do not come to class, then there will not be a quiz.</td></tr>
<tr><td>5. If there is not going to be a quiz, then I don’t come to class.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$ and $C = \{4, 5, 6, 7, 8, 9, 10\}$.
Find $A \cap B \cap C$.</td><td>1. {4, 6}</td></tr>
<tr><td>2. {0, 2, 4, 6}</td></tr>
<tr><td>3. {4, 5, 6}</td></tr>
<tr><td>4. {4, 5, 6, 7 , 8, 9, 10}</td></tr>
<tr><td>5. $\varnothing$</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of three-digit numbers made from the digits 6, 7, 8 and 9. How many numbers in S have no repeated digit?</td><td>1. 81</td></tr>
<tr><td>2. 4</td></tr>
<tr><td>3. 24</td></tr>
<tr><td>4. 1000</td></tr>
<tr><td>5. 27</td></tr>
<tr><td rowspan="5">Assignment 4
Find the sum-of-products expansion of the Boolean function $f(x, y, z)$, which is given in the table:
<table><tr><th>x</th><th>y</th><th>z</th><th>$f(x, y, z)$</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table></td><td>1. $x \wedge \bar{y} \wedge z$</td></tr>
<tr><td>2. $(\bar{x} \wedge y \wedge \bar{z}) \vee (x \wedge \bar{y} \wedge z)$</td></tr>
<tr><td>3. $\bar{x} \vee y \vee \bar{z}$</td></tr>
<tr><td>4. $x \vee \bar{y} \vee z$</td></tr>
<tr><td>5. $(x \wedge y \wedge z) \vee (x \wedge \bar{y} \wedge \bar{z})$</td></tr>
<tr><td rowspan="5">Assignment 5
Simplify the sum-of-products expansion of the Boolean function represented by the following K-map:
<table><tr><td></td><td>x</td><td>$\bar{x}$</td></tr><tr><td>y</td><td>1</td><td>1</td></tr><tr><td>$\bar{y}$</td><td>1</td><td></td></tr></table></td><td>1. $\bar{x} \vee \bar{y}$</td></tr>
<tr><td>2. y</td></tr>
<tr><td>3. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td>4. x</td></tr>
<tr><td>5. $x \vee y$</td></tr>
</table>

<table>
<tr><td>Assignment 6
Let A $= \{a, b\}$. Which of the following relations on a set A are symmetric?
$S = \{(a,a),(a,b),(b,a)\}$,
$T = \{(a,a),(a,b),(b,b)\}$,
$U = \{(a,b),(b,a)\}$,
$V = \{(b,b),(a,b),(b,a)\}$.</td><td>1. S, T, U, V
2. S, U, V
3. T
4. S, T
5. T, U</td></tr>
<tr><td>Assignment 7
Write down the left child of vertex c in the binary tree in the figure:
 
</td><td>1. a
2. d
3. e
4. f
5. g</td></tr>
<tr><td>Assignment 8
Use the coding scheme
<table><tr><th>Символ</th><th>Код</th></tr><tr><td>a</td><td>111</td></tr><tr><td>c</td><td>101</td></tr><tr><td>e</td><td>01</td></tr><tr><td>i</td><td>11011</td></tr><tr><td>g</td><td>0010</td></tr><tr><td>h</td><td>1001</td></tr><tr><td>m</td><td>0001</td></tr><tr><td>s</td><td>10001</td></tr><tr><td>t</td><td>11001</td></tr></table>to decode the word 11001100101000101.</td><td>1. $mathematics$
2. $gates$
3. $seismic$
4. $theme$
5. $agate$</td></tr>
<tr><td>Assignment 9
Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two bs.</td><td>1. $a^*ba^*ba^*$
2. $b^*ab^*ab^*$
3. $(a \cup c)^*b(a \cup c)^*b(a \cup c)^*$
4. $(b \cup c)^*a(b \cup c)^*a$
5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a$</td></tr>
</table>

Assignment 10

Which of the following words are not accepted by the following automaton?

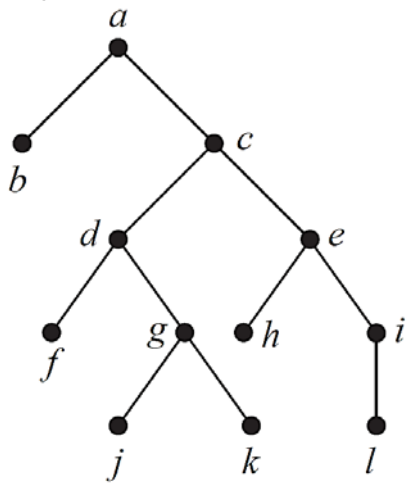


1. aab
2. $aabb$
3. abb
4. $aabbb$
5. aa

Assignment 11

Eulerian graphs.

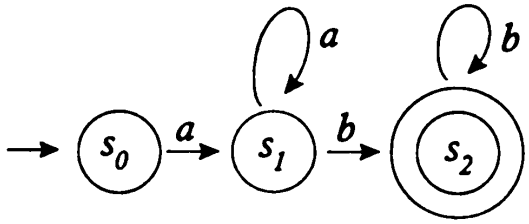
<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the contrapositive of this conditional statement. I come to class whenever there is going to be a quiz.</td><td>1. If I do not come to class, then there will be a quiz.</td></tr>
<tr><td>2. If I come to class, then there will be a quiz.</td></tr>
<tr><td>3. If I come to class, then there will not be a quiz.</td></tr>
<tr><td>4. If I do not come to class, then there will not be a quiz.</td></tr>
<tr><td>5. If there is not going to be a quiz, then I don’t come to class.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{0, 2, 4\}, B = \{0, 1, 2, 3\}$ and $C = \{4, 5\}$. Find $A \cup B \cup C$.</td><td>1. {0, 1, 2, 3, 4, 5}</td></tr>
<tr><td>2. {0, 2, 4, 5}</td></tr>
<tr><td>3. {0, 1, 2, 3, 4}</td></tr>
<tr><td>4. {0, 1, 2, 3, 5}</td></tr>
<tr><td>5. $\varnothing$</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of three-digit numbers made from the digits 1, 6, 7, 8 and 9. How many numbers in S have no repeated digit?</td><td>1. 5</td></tr>
<tr><td>2. 125</td></tr>
<tr><td>3. 1000</td></tr>
<tr><td>4. 75</td></tr>
<tr><td>5. 60</td></tr>
<tr><td rowspan="5">Assignment 4
Find the sum-of-products expansion of the Boolean function $f(x, y, z)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>z</th><th>$f(x, y, z)$</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table></td><td>1. $x \wedge y \wedge z$</td></tr>
<tr><td>2. $(\bar{x} \wedge y \wedge \bar{z}) \vee (x \wedge \bar{y} \wedge z)$</td></tr>
<tr><td>3. $\bar{x} \vee y \vee \bar{z}$</td></tr>
<tr><td>4. $x \vee \bar{y} \vee \bar{z}$</td></tr>
<tr><td>5. $(x \wedge y \wedge z) \vee (x \wedge \bar{y} \wedge \bar{z})$</td></tr>
<tr><td rowspan="5">Assignment 5
Simplify the sum-of-products expansion of the Boolean function represented by the following K-map:
<table><tr><th></th><th>x</th><th>$\bar{x}$</th></tr><tr><td>y</td><td>1</td><td>1</td></tr><tr><td>$\bar{y}$</td><td></td><td>1</td></tr></table></td><td>1. $\bar{x} \vee y$</td></tr>
<tr><td>2. y</td></tr>
<tr><td>3. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td>4. x</td></tr>
<tr><td>5. $x \vee y$</td></tr>
</table>

Assignment 6 The characteristic equation of a linear homogeneous recurrence relation with constant coefficients (for a sequence $\{a_n\}$) is of the form: $(r-5)\cdot(r-2)$. What is the general solution of this recurrence relation?	1. $a_n = C_1 \cdot 2^n + C_2 \cdot 5^n$ 2. $a_n = C \cdot 10^n$ 3. $a_n = C \cdot 2^n$ 4. $a_n = C \cdot 5^n$ 5. $a_n = 10 \cdot (2^n + 5^n)$
Assignment 7 Write down the right child of vertex c in the following binary tree:  	1. a 2. d 3. e 4. f 5. g
Assignment 8 Use the coding scheme to decode the word 11100101111100101.	1. $mathematics$ 2. $gates$ 3. $seismic$ 4. $theme$ 5. $agate$
Assignment 9 Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two as.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b$ 4. $(b \cup c)^*a(b \cup c)^*a(b \cup c)^*$ 5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a(a \cup b \cup c)^*$

Символ	Код
a	111
c	101
e	01
i	11011
g	0010
h	1001
m	0001
s	10001
t	11001

Assignment 10

Which of the following words are accepted by the following automaton?

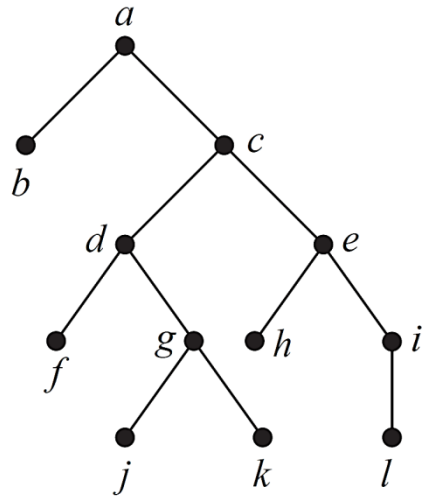
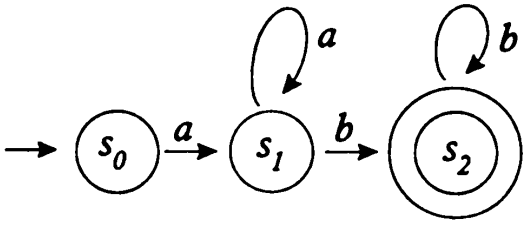


1. aa
2. aaa
3. abb
4. $aaaa$
5. ba

Assignment 11

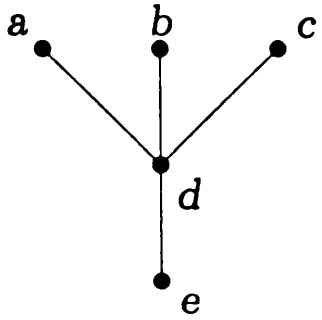
Hamiltonian graphs.

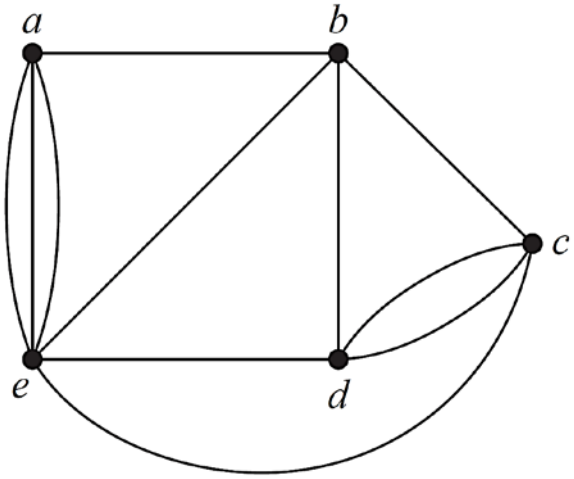
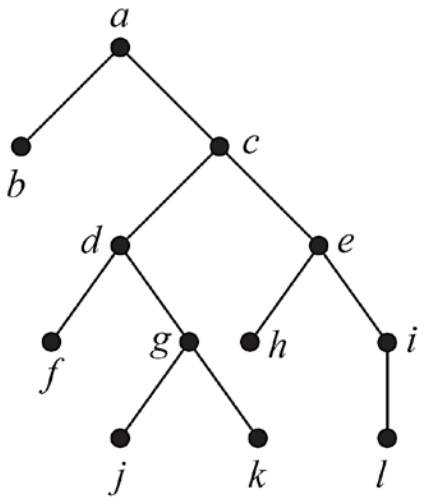
<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td>Assignment 1
Suppose that the domain of the propositional function $P(x)$ consists of the integers 1, 2 and 3. Write out the proposition $\exists x P(x)$ using conjunctions ($\wedge$), disjunctions ($\vee$) and negations ($^{-}$).</td><td>1. $P(1) \vee P(2) \vee P(3)$</td></tr>
<tr><td></td><td>2. $(P(1) \wedge P(2)) \vee P(3)$</td></tr>
<tr><td></td><td>3. $P(1) \wedge P(2) \wedge P(3)$</td></tr>
<tr><td></td><td>4. $\overline{P(1)} \wedge \overline{P(2)} \wedge \overline{P(3)}$</td></tr>
<tr><td></td><td>5. $\overline{P(1)} \vee \overline{P(2)} \vee \overline{P(3)}$</td></tr>
<tr><td>Assignment 2
Let $A = \{a\}$ and $B = \{y, z\}$. Find $A \times B$.</td><td>1. $(a, y), (a, z)$</td></tr>
<tr><td></td><td>2. a, y, z</td></tr>
<tr><td></td><td>3. $(a, a), (y, y), (z, z)$</td></tr>
<tr><td></td><td>4. $(a, y), z$</td></tr>
<tr><td></td><td>5. $(a, a), (y, z), (z, y)$</td></tr>
<tr><td>Assignment 3
Find the coefficient of $a^3 b^5$ in the expansion $(a + b)^8$.</td><td>1. 78</td></tr>
<tr><td></td><td>2. 56</td></tr>
<tr><td></td><td>3. 8</td></tr>
<tr><td></td><td>4. 1</td></tr>
<tr><td></td><td>5. 4</td></tr>
<tr><td>Assignment 4
Find the product-of-sums expansion of the Boolean function $f(x, y, z)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>z</th><th>$f(x, y, z)$</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table></td><td>1. $x \vee \bar{y} \vee z$</td></tr>
<tr><td></td><td>2. $(\bar{x} \wedge y \wedge \bar{z}) \vee (x \wedge \bar{y} \wedge z)$</td></tr>
<tr><td></td><td>3. $\bar{x} \vee y \vee \bar{z}$</td></tr>
<tr><td></td><td>4. $(x \vee \bar{y} \vee z) \wedge (\bar{x} \vee y \vee \bar{z})$</td></tr>
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<table><tr><td></td><td>x</td><td>$\bar{x}$</td></tr><tr><td>y</td><td></td><td>1</td></tr><tr><td>$\bar{y}$</td><td>1</td><td>1</td></tr></table></td><td>1. $\bar{x} \vee \bar{y}$</td></tr>
<tr><td></td><td>2. y</td></tr>
<tr><td></td><td>3. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td></td><td>4. x</td></tr>
<tr><td></td><td>5. $x \vee y$</td></tr>
<tr><td>Assignment 6
Let A $= \{a, b\}$. Which of the following</td><td>1. S, T, U, V</td></tr>
<tr><td></td><td>2. S, U, V</td></tr>
</table>

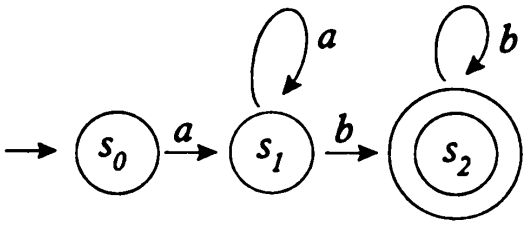
relations on a set A are reflexive and transitive? $S = \{(a,a),(a,b),(b,a)\}$, $T = \{(a,a),(a,b),(b,b)\}$, $U = \{(a,b),(b,a)\}$, $V = \{(b,b),(a,b),(b,a)\}$.	3. T 4. S 5. T, U
Assignment 7 Determine the depth of the rooted tree in the figure:  	1. 2 2. 3 3. 4 4. 5 5. 11
Assignment 8 An encoding function is defined by the generator matrix $G = \begin{bmatrix} 1\,0\,0\,1\,1\,1 \\ 0\,1\,0\,1\,0\,1 \\ 0\,0\,1\,1\,1\,1 \end{bmatrix}$. Encode 011.	1. 011010 2. 100111 3. 011101 4. 100110 5. 111100
Assignment 9 Let $A = \{a,b,c\}$. Give a regular expression for the set of all elements of A^*, containing exactly two bs.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b(a \cup c)^*$ 4. $(b \cup c)^*a(b \cup c)^*a(b \cup c)^*$ 5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a(a \cup b \cup c)^*$
Assignment 10 Find a regular expression for the language accepted by the automaton:  	1. $a^*ba^*ba^*$ 2. $ab(aab)^*$ 3. aa^*bb^* 4. a^*ba^* 5. $a^*ba^*ba^*b$

Assignment 11

Spanning tree in a graph.

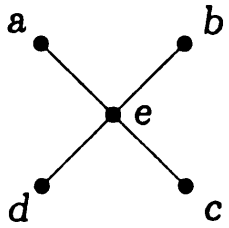
Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x, y)$ consists of pairs (x, y), where $x = 1, 2$ and $y = 0, 2, 3$. Write out the proposition $\forall x P(x, 3)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(1,3) \vee P(2,3)$ 2. $P(1,3) \wedge P(2,3)$ 3. $P(2,2) \vee P(2,3)$ 4. $P(2,0) \vee P(2,3)$ 5. $P(2,0) \wedge P(2,2)$
Assignment 2 List all elements incomparable with d in the partial ordering with the accompanying Hasse diagram.  	1. a, b, c 2. e 3. d 4. a, b, c, e 5. Such elements don't exist.
Assignment 3 A sequence of integers is defined by $f(0) = 1$, $f(1) = 3, f(k) = 2 \cdot f(k-1) - f(k-2)$. Calculate $f(3)$.	1. 1 2. 3 3. 7 4. 16 5. 5
Assignment 4 The Boolean operator $\oplus$, called the XOR operator, is defined by $x \oplus y = (\bar{x} \wedge y) \vee (x \wedge \bar{y})$. Simplify the expression $x \oplus x$.	1. 0 2. 1 3. x 4. $\bar{x}$ 5. $x \oplus x$
Assignment 5 Simplify the Boolean function $f(x, y, z, w) = (x \wedge y \wedge z \wedge w) \vee (\bar{x} \wedge y \wedge z \wedge w)$.	1. $y \wedge z \wedge w$ 2. $x \wedge z \wedge w$ 3. $x \wedge y \wedge w$ 4. $x \wedge y \wedge z$ 5. $\bar{x} \wedge y \wedge z \wedge w$
Assignment 6 Find an Eulerian path in the Eulerian graph in the figure:	1. $edbae$ 2. $edbe$ 3. $edcdbe$ 4. $edbaeae$ 5. $edbaeaebcdce$

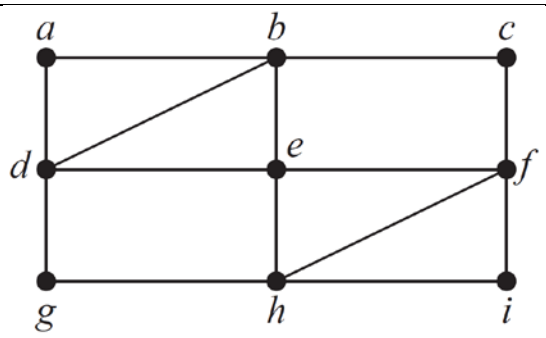
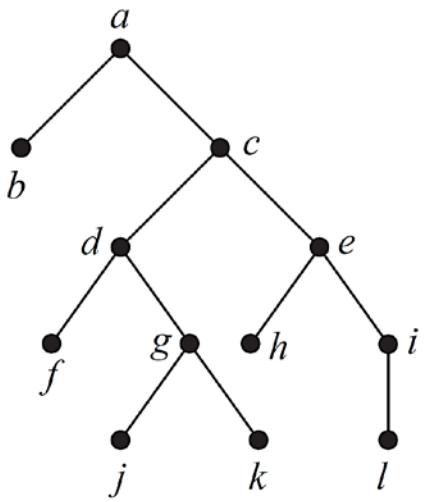
	
Assignment 7 Write down all leaves in the following binary rooted tree: 	1. e, g, h, i, j, k 2. b, c 3. b, f, j, k, h, l 4. d, e, f, g 5. e, g, h, i, j, k
Assignment 8 The minimum distance of the code is 3. Find the maximum number of errors which the code can detect.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Find a regular expression corresponding to the regular set $\{ba, ca, da\}$.	1. $(a \vee b)(b \vee c)$ 2. $a(\lambda \vee c \vee d)b$ 3. $(a \vee b \vee c)(d \vee e \vee f)$ 4. $(b \vee c \vee d)a$ 5. $a(b \vee c \vee d)$
Assignment 10 Find a regular expression for the language accepted by the automaton:	1. $a^*ba^*ba^*$ 2. $ab(aab)^*$ 3. aa^*bb^* 4. a^*ba^* 5. $a^*ba^*ba^*b$

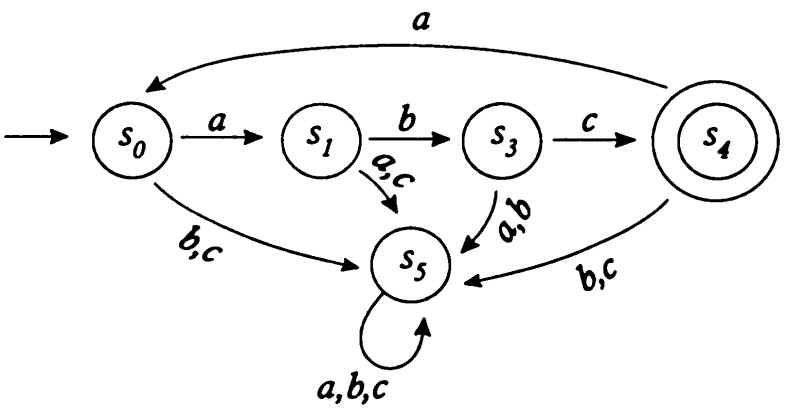


Assignment 11

Planar graphs.

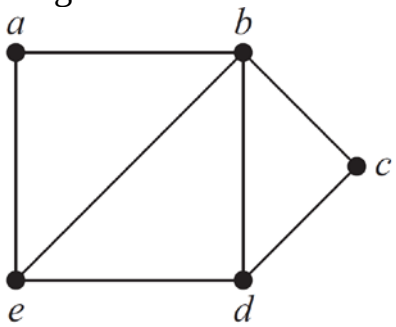
Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x, y)$ consists of pairs (x, y), where $x = 1, 2$ and $y = 0, 2, 3$. Write out the proposition $\exists x P(x, 3)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(1,3) \vee P(2,3)$ 2. $P(1,3) \wedge P(2,3)$ 3. $P(2,2) \vee P(2,3)$ 4. $P(2,0) \vee P(2,3)$ 5. $P(2,0) \wedge P(2,2)$
Assignment 2 List all elements incomparable with a in the partial ordering with the accompanying Hasse diagram.  	1. e, b, c 2. b 3. d, c 4. c, d, b 5. Such elements don't exist.
Assignment 3 A sequence of integers is defined by $f(0) = 2$, $f(1) = 4, f(k) = 3 \cdot f(k-1) - 2 \cdot f(k-2)$. Calculate $f(3)$.	1. 24 2. 8 3. 16 4. 4 5. 6
Assignment 4 The Boolean operator $\oplus$, called the XOR operator, is defined by $x \oplus y = (\bar{x} \wedge y) \vee (x \wedge \bar{y})$. Simplify the expression $x \oplus \bar{x}$.	1. 0 2. 1 3. x 4. $\bar{x}$ 5. $x \oplus \bar{x}$
Assignment 5 Simplify the Boolean function $f(x, y, z, w) = (x \wedge y \wedge z \wedge w) \vee (x \wedge \bar{y} \wedge z \wedge w)$.	1. $y \wedge z \wedge w$ 2. $x \wedge z \wedge w$ 3. $x \wedge y \wedge w$ 4. $x \wedge y \wedge z$ 5. $x \wedge \bar{y} \wedge z \wedge w$
Assignment 6 Find an Eulerian path in the Eulerian graph in the figure:	1. $fihefhef$ 2. $fedbehfcbadghif$ 3. $fcbadghif$ 4. $fedbehf$ 5. $fcbef$

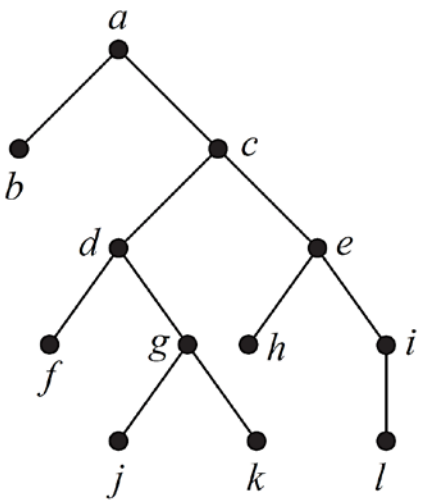
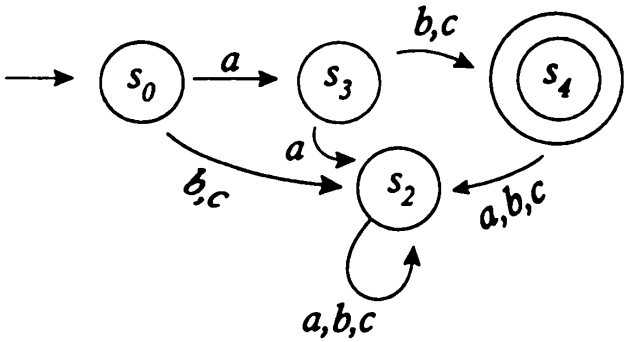
	
Assignment 7 Write down all internal vertices of the following rooted tree 	1. d, f, g, h, j, k 2. b, c, e, i, d, g 3. c, e, i, d, g 4. d, f, g, j, k 5. b, c, d, f
Assignment 8 The minimum distance of the code is 3. Find the maximum number of errors which the code can correct.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Find a regular expression corresponding to the regular set $\{ab, ac, ad, bb, bc, bd\}$.	1. $(a \vee b)(b \vee c)$ 2. $a(\lambda \vee c \vee d)b$ 3. $(a \vee b \vee c)(d \vee e \vee f)$ 4. $(b \vee c \vee d)a$ 5. $(a \vee b)(b \vee c \vee d)$
Assignment 10 Find a regular expression for the language accepted by the automaton:	1. $a^*ba^*ba^*$ 2. $abc(aabc)^*$ 3. $a(b \vee c)$ 4. a^*ba^*c 5. $a^*ba^*ba^*c$

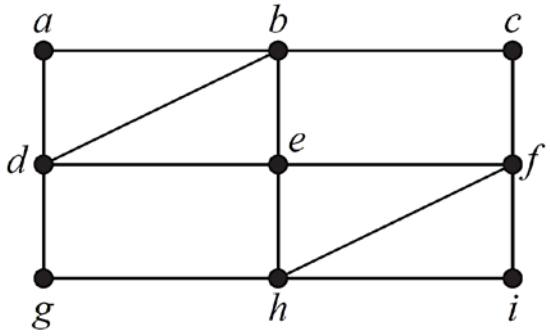


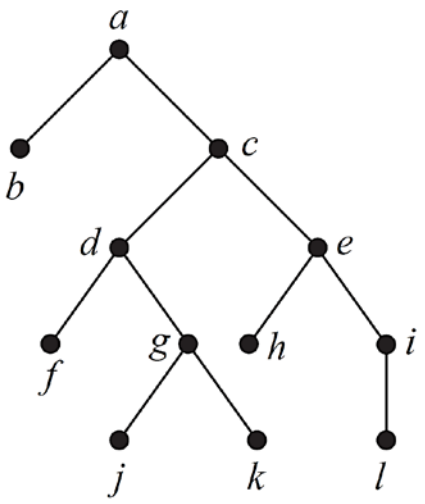
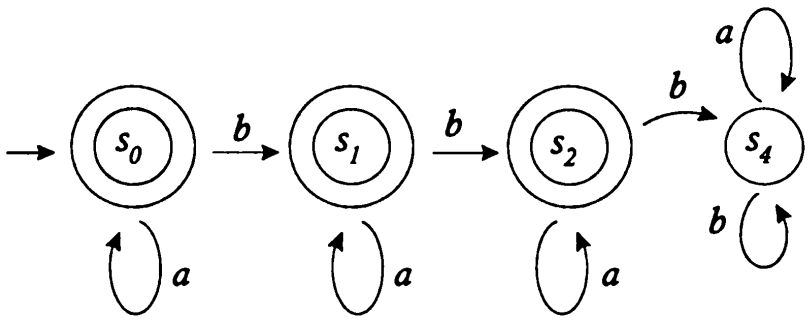
Assignment 11

Euler's formula for connected planar graphs.

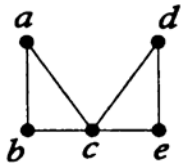
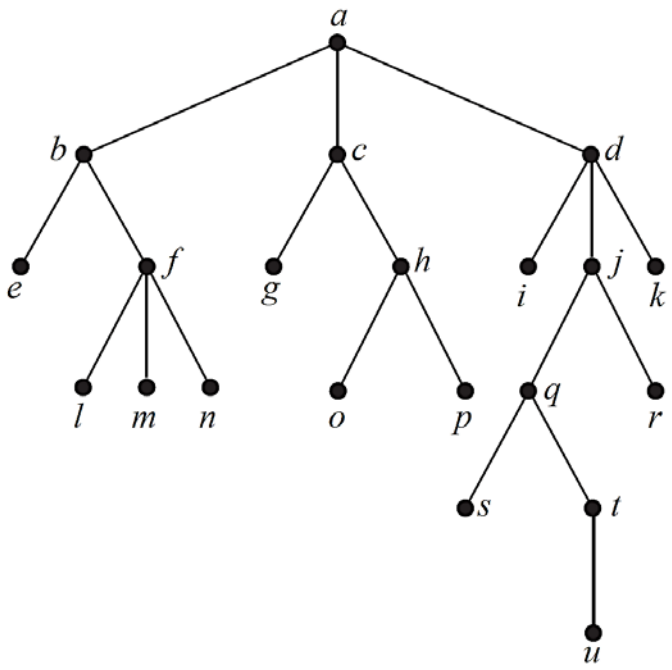
Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x,y)$ consists of pairs (x,y), where $x=1,2$ and $y=0,2,3$. Write out the proposition $\forall y P(2,y)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(2,0)\vee P(2,2)\vee P(2,3)$ 2. $P(2,0)\wedge P(2,2)\wedge P(2,3)$ 3. $P(2,2)\vee P(2,3)$ 4. $P(2,0)\vee P(2,3)$ 5. $P(2,0)\wedge P(2,2)$
Assignment 2 Suppose that $A=\{1,2,3,4,5,6,7,8,9,10\}$. Let R be the relation on the set A such that $(x,y)\in R$ if and only if $x-y$ is an even integer. R is an equivalence relation. What is the equivalence class [3] for this equivalence relation?	1. $\{2,4,6,8,10\}$ 2. $\{1,3,5,7,9\}$ 3. $\{6\}$ 4. $\{3,6,9\}$ 5. $\emptyset$
Assignment 3 Find the number of elements in the union $A_1\cup A_2\cup A_3$ if there are 100 elements in each set A_1,A_2,A_3 and these sets are pairwise disjoint.	1. 100 2. 300 3. 225 4. 200 5. 150
Assignment 4 The Boolean operator | , called the NAND operator, is defined by $x\mid y=\overline{x\wedge y}$. Express $\bar{x}$ using just the | operator.	1. $x\mid(x\mid x)$ 2. $\bar{x}\mid x$ 3. $x\mid x$ 4. $x\mid(\bar{x}\mid\bar{x})$ 5. $\bar{x}\mid(x\mid x)$
Assignment 5 Simplify the Boolean function $f(x,y,z,w)=(x\wedge y\wedge z\wedge w)\vee(x\wedge y\wedge\bar{z}\wedge w)$.	1. $y\wedge z\wedge w$ 2. $x\wedge z\wedge w$ 3. $x\wedge y\wedge w$ 4. $x\wedge y\wedge z$ 5. $x\wedge y\wedge\bar{z}\wedge w$
Assignment 6 Find a Hamiltonian cycle in the Hamiltonian graph in the figure:  	1. $ebcde$ 2. $ebcdbe$ 3. $abcdea$ 4. $ebde$ 5. $eabde$

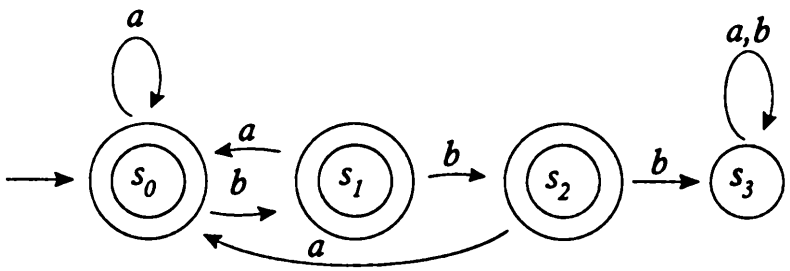
Assignment 7 Write down the left child of vertex d in the following binary tree:  	1. b 2. c 3. e 4. f 5. g
Assignment 8 The minimum distance of the code is 4. Find the maximum number of errors which the code can detect.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two bs.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b(a \cup c)^*$ 4. $(b \cup c)^*a(b \cup c)^*a$ 5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a$
Assignment 10 Find a regular expression for the language accepted by the automaton:  	1. $a^*ba^*ba^*$ 2. $abc(aabc)^*$ 3. $a(b \vee c)$ 4. a^*ba^*c 5. $a^*ba^*ba^*c$
Assignment 11 An algorithm which produces a minimal spanning tree (Prim's algorithm).	

Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x,y)$ consists of pairs (x,y), where $x=1,2$ and $y=0,2,3$. Write out the proposition $\exists y P(2,y)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(2,0)\vee P(2,2)\vee P(2,3)$ 2. $P(2,0)\wedge P(2,2)\wedge P(2,3)$ 3. $P(2,2)\vee P(2,3)$ 4. $P(2,0)\vee P(2,3)$ 5. $P(2,0)\wedge P(2,2)$
Assignment 2 Suppose that $A=\{1,2,3,4,5,6,7,8,9,10\}$. Let R be the relation on the set A such that $(x,y)\in R$ if and only if $x-y$ is an even integer. R is an equivalence relation. What is the equivalence class $[6]$ for this equivalence relation?	1. $\{2,4,6,8,10\}$ 2. $\{1,3,5,7,9\}$ 3. $\{6\}$ 4. $\{3,6,9\}$ 5. $\emptyset$
Assignment 3 There are x computer science students at a university. Of these, 94 have taken a course in $C^{\#}$, 70 have taken a course in C^{++} and 81 have taken a course in $Java$. Each computer science student has taken a course only in one programming language. Find x.	1. 151 2. 175 3. 245 4. 164 5. 140
Assignment 4 The Boolean operator $\downarrow$, called the NOR operator, is defined by $x\downarrow y=\overline{x\vee y}$. Express $\bar{x}$ using just the $\downarrow$ operator.	1. $x\downarrow(x\downarrow x)$ 2. $\bar{x}\downarrow x$ 3. $x\downarrow x$ 4. $x\downarrow(\bar{x}\downarrow\bar{x})$ 5. $\bar{x}\downarrow(x\downarrow x)$
Assignment 5 Simplify the Boolean function $f(x,y,z,w)=(x\wedge y\wedge z\wedge w)\vee(x\wedge y\wedge z\wedge\bar{w})$.	1. $y\wedge z\wedge w$ 2. $x\wedge z\wedge w$ 3. $x\wedge y\wedge w$ 4. $x\wedge y\wedge z$ 5. $x\wedge y\wedge z\wedge\bar{w}$
Assignment 6 Find a Hamiltonian cycle in the Hamiltonian graph in the figure: 	1. $ecbe$ 2. $ebadghifce$ 3. $ecbae$ 4. $ecbaeighe$ 5. $efcbade$

Assignment 7 Write down the right child of vertex d in the following binary tree:  	1. a 2. c 3. e 4. f 5. g
Assignment 8 The minimum distance of the code is 5. Find the maximum number of errors which the code can correct.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two as.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b$ 4. $(b \cup c)^*a(b \cup c)^*a(b \cup c)^*$ 5. $(a \cup b \cup c)^*aa(a \cup b \cup c)^*$
Assignment 10 Find a regular expression for the language accepted by the automaton:  	1. $a^*ba^*ba^*$ 2. $abc(aabc)^*$ 3. $a(b \vee c)$ 4. a^*ba^*c 5. $a^*ba^*ba^*c$
Assignment 11 Dijkstra's algorithm.	

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the converse of this conditional statement. If it snows today, I will sky tomorrow.</td><td>1. I will sky tomorrow only if it snows today.</td></tr>
<tr><td>2. If I do not sky tomorrow, then it will not have snowed today.</td></tr>
<tr><td>3. If it does not snow today, then I will not sky tomorrow.</td></tr>
<tr><td>4. If it snows today, then I will not sky tomorrow.</td></tr>
<tr><td>5. If it does not snow today, then I will sky tomorrow.</td></tr>
<tr><td rowspan="5">Assignment 2
A palindrome is a string of digits that reads the same backwards as forwards. How many different palindromes are there with six digits?</td><td>1. 900</td></tr>
<tr><td>2. 1000</td></tr>
<tr><td>3. 9000</td></tr>
<tr><td>4. 850</td></tr>
<tr><td>5. 10000</td></tr>
<tr><td rowspan="5">Assignment 3
Each of the x first-year students studying computing at Yanka Kupala Grodno State University can study a number of optional units. Of these, 25 chose to study the accounting option, 13 chose the business option and 8 studied both of these options/ Find x.</td><td>1. 38</td></tr>
<tr><td>2. 30</td></tr>
<tr><td>3. 28</td></tr>
<tr><td>4. 17</td></tr>
<tr><td>5. 27</td></tr>
<tr><td rowspan="5">Assignment 4
Find the sum-of-products expansion of the Boolean function $f(x, y)$ that has the value 1 if and only if $x = 1$, $y = 0$.</td><td>1. $x \wedge \bar{y}$</td></tr>
<tr><td>2. $\bar{x} \wedge y$</td></tr>
<tr><td>3. $x \vee \bar{y}$</td></tr>
<tr><td>4. $\bar{x} \vee y$</td></tr>
<tr><td>5. $x \wedge \bar{y}$</td></tr>
<tr><td rowspan="5">Assignment 5
Simplify the sum-of-products expansion of the Boolean function represented by the following K-map:
<table><tr><td></td><td>y</td><td>y</td><td>$\bar{y}$</td><td>$\bar{y}$</td><td></td></tr><tr><td>x</td><td>1</td><td>1</td><td></td><td></td><td>w</td></tr><tr><td>x</td><td>1</td><td>1</td><td></td><td></td><td>$\bar{w}$</td></tr><tr><td>$\bar{x}$</td><td>1</td><td>1</td><td></td><td></td><td>$\bar{w}$</td></tr><tr><td>$\bar{x}$</td><td>1</td><td>1</td><td></td><td></td><td>w</td></tr><tr><td></td><td>z</td><td>$\bar{z}$</td><td>$\bar{z}$</td><td>z</td><td></td></tr></table></td><td>1. $x \wedge y$</td></tr>
<tr><td>2. x</td></tr>
<tr><td>3. z</td></tr>
<tr><td>4. w</td></tr>
<tr><td>5. y</td></tr>
</table>

Assignment 6 Determine all edges of the following graph incident to vertex a:  	1. $\{a,d\},\{a,c\}$ 2. $\{c,b\},\{c,d\},\{e,d\}$ 3. $\{c,d\},\{a,c\}$ 4. $\{a,b\},\{a,c\}$ 5. Such edges don't exist.
Assignment 7 Determine the depth of the rooted tree in the figure:  	1. 6 2. 3 3. 4 4. 5 5. 9
Assignment 8 An encoding function is defined by the generator matrix $G = \begin{bmatrix} 1\,0\,0\,1\,1\,0 \\ 0\,1\,0\,1\,0\,1 \\ 0\,0\,1\,0\,0\,0 \end{bmatrix}$. Encode 100.	1. 011010 2. 100111 3. 011101 4. 100110 5. 111100
Assignment 9 Find a regular set corresponding to the regular expression $a(b \vee c)d$.	1. $\{aba, aca, ada\}$ 2. $\{abd, acd\}$ 3. $\{ac, ad, bc, bd\}$ 4. $\{acb, adb, bcb, bdb\}$ 5. $\{aac, aad, abc, abd\}$
Assignment 10 Determine which language is accepted by the automaton:	1. All words which do not have three consecutive bs. 2. All words which have three consecutive bs. 3. All words which do



not have three consecutive as.

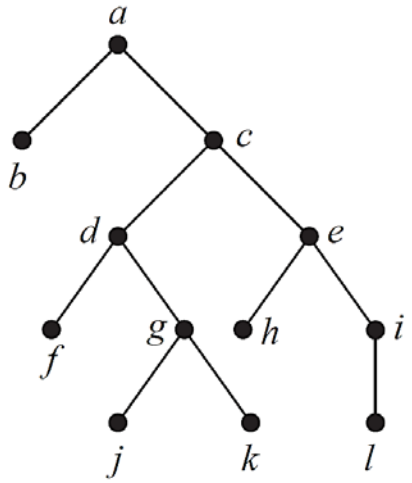
4. All words which have three consecutive as.

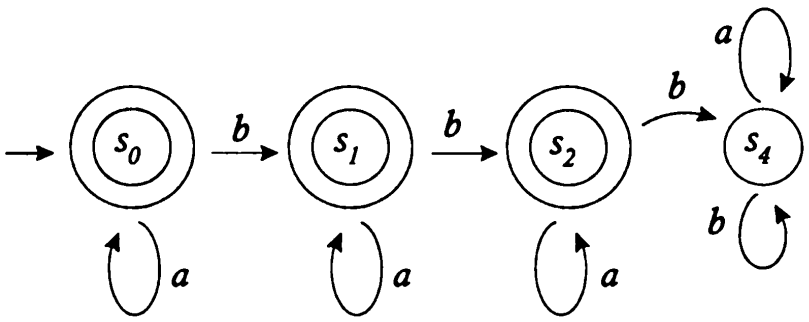
5. All words which have four consecutive bs.

Assignment 11

The nearest neighbour algorithm

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the contrapositive of this conditional statement. If it snows today, I will sky tomorrow.</td><td>1. I will sky tomorrow only if it snows today.</td></tr>
<tr><td>2. If I do not sky tomorrow, then it will not have snowed today.</td></tr>
<tr><td>3. If it does not snow today, then I will not sky tomorrow.</td></tr>
<tr><td>4. If it snows today, then I will not sky tomorrow.</td></tr>
<tr><td>5. If it does not snow today, then I will sky tomorrow.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{a, b, c, d\}$ and $B = \{a, b, e, f, g, h\}$. Find $B \setminus A$.</td><td>1. $\{e, f, g, h\}$</td></tr>
<tr><td>2. $\{f, g, h\}$</td></tr>
<tr><td>3. $\{c, d\}$</td></tr>
<tr><td>4. $\varnothing$</td></tr>
<tr><td>5. $\{a, b, c, g, h\}$</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of three-digit numbers made from the digits 1, 2, 3, 6 and 8. How many numbers are there in S?</td><td>1. 125</td></tr>
<tr><td>2. 5</td></tr>
<tr><td>3. 50</td></tr>
<tr><td>4. 15</td></tr>
<tr><td>5. 300</td></tr>
<tr><td rowspan="5">Assignment 4
Find the product-of-sums expansion of the Boolean function $f(x, y)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>$f(x, y)$</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table></td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge y)$</td></tr>
<tr><td>2. $(\bar{x} \wedge y) \vee (x \wedge \bar{y})$</td></tr>
<tr><td>3. $(x \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>4. $(\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>5. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td rowspan="5">Assignment 5
Find the dual of the Boolean function $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 0)$.</td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 1)$</td></tr>
<tr><td>2. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 0)$</td></tr>
<tr><td>3. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 0)$</td></tr>
<tr><td>4. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 1)$</td></tr>
<tr><td>5. $(\bar{x} \vee \bar{y}) \vee (\bar{x} \vee 0)$</td></tr>
<tr><td rowspan="5">Assignment 6
Let A $= \{a, b\}$. Which of the following relations on a set A are reflexive?
$S = \{(a, a), (a, b), (b, a)\}$,
$T = \{(a, a), (a, b), (b, b)\}$,
$U = \{(a, b), (b, a)\}$,
$V = \{(b, b), (a, b), (b, a)\}$.</td><td>1. S, T, U, V</td></tr>
<tr><td>2. T</td></tr>
<tr><td>3. U</td></tr>
<tr><td>4. S, U</td></tr>
<tr><td>5. T, U</td></tr>
</table>

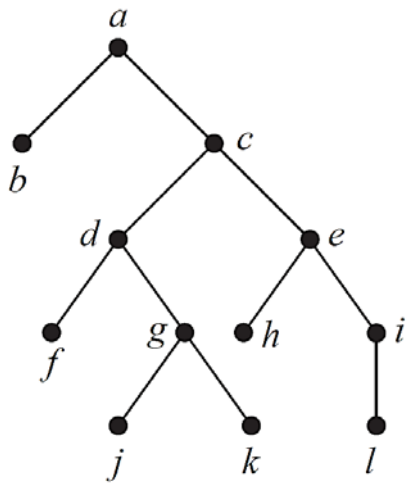
<table>
<tr>
<td>Assignment 7
Write down all leaves of the rooted tree in the figure:
 
</td>
<td>1. b, f, j, h, l
2. d, f, i
3. b, f, j, h, l, k
4. d, f, g
5. b, c, e, i</td>
</tr>
<tr>
<td>Assignment 8
Use the coding scheme
<table><tr><th>Символ</th><th>Код</th></tr><tr><td>a</td><td>111</td></tr><tr><td>c</td><td>101</td></tr><tr><td>e</td><td>01</td></tr><tr><td>i</td><td>11011</td></tr><tr><td>g</td><td>0010</td></tr><tr><td>h</td><td>1001</td></tr><tr><td>m</td><td>0001</td></tr><tr><td>s</td><td>10001</td></tr><tr><td>l</td><td>11001</td></tr></table>to encode the word $mammals$.</td>
<td>1. 11110110111100110000100110001
2. 1111101010110000110001
3. 10101011100101010101
4. 1010111001010001010
5. 000111100010001111110011000 1</td>
</tr>
<tr>
<td>Assignment 9
Find a regular expression corresponding to the regular set $\{ba, ca, da\}$.</td>
<td>1. $(a \vee b)(b \vee c)$
2. $a(\lambda \vee c \vee d)b$
3. $(a \vee b \vee c)(d \vee e \vee f)$
4. $(b \vee c \vee d)a$
5. $a(b \vee c \vee d)$</td>
</tr>
<tr>
<td>Assignment 10
Which of the following words are not accepted by the following automaton?</td>
<td>1. b
2. bb
3. bba
4. $bbba$
5. aa</td>
</tr>
</table>

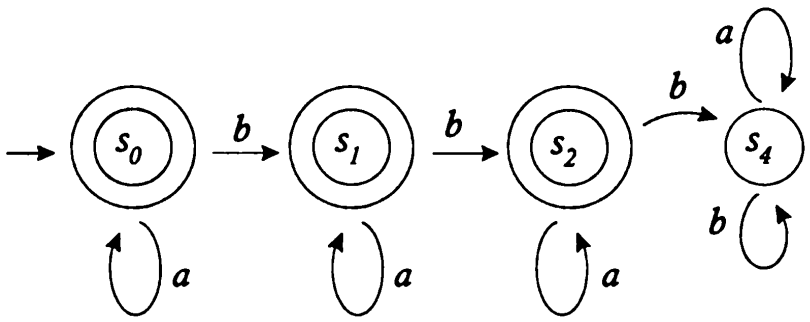


Assignment 11

The nearest insertion algorithm.

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the inverse of this conditional statement. If it snows today, I will sky tomorrow.</td><td>1. I will sky tomorrow only if it snows today.</td></tr>
<tr><td>2. If I do not sky tomorrow, then it will not have snowed today.</td></tr>
<tr><td>3. If it does not snow today, then I will not sky tomorrow.</td></tr>
<tr><td>4. If it snows today, then I will not sky tomorrow.</td></tr>
<tr><td>5. If it does not snow today, then I will sky tomorrow.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find $B \setminus A$.</td><td>1.{1, 2, 4, 5}</td></tr>
<tr><td>2.{0, 6}</td></tr>
<tr><td>3.{0, 1, 2, 3, 4, 5, 6}</td></tr>
<tr><td>4.{3, 6}</td></tr>
<tr><td>5.{3}</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of four-digit numbers made from the digits 6, 7 and 9. How many numbers are there in S?</td><td>1. 27</td></tr>
<tr><td>2. 81</td></tr>
<tr><td>3. 3</td></tr>
<tr><td>4. 15</td></tr>
<tr><td>5. 9</td></tr>
<tr><td rowspan="5">Assignment 4
Find the product-of-sums expansion of the Boolean function $f(x, y)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>$f(x,y)$</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table></td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge y)$</td></tr>
<tr><td>2. $(\bar{x} \wedge y) \vee (x \wedge \bar{y})$</td></tr>
<tr><td>3. $(x \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>4. $(\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y})$</td></tr>
<tr><td>5. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td rowspan="5">Assignment 5
Find the dual of the Boolean function $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 1)$.</td><td>1. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 1)$</td></tr>
<tr><td>2. $(\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge 0)$</td></tr>
<tr><td>3. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 0)$</td></tr>
<tr><td>4. $(\bar{x} \vee \bar{y}) \wedge (\bar{x} \vee 1)$</td></tr>
<tr><td>5. $(\bar{x} \vee \bar{y}) \vee (\bar{x} \vee 1)$</td></tr>
<tr><td rowspan="5">Assignment 6
The characteristic equation of a linear homogeneous recurrence relation with constant coefficients (for a sequence $\{a_n\}$) is of the form: $(r - 2) \cdot (r - 3)$.
What is the general solution of this recurrence relation?</td><td>1. $a_n = C \cdot 6^n$</td></tr>
<tr><td>2. $a_n = C_1 \cdot 2^n + C_2 \cdot 3^n$</td></tr>
<tr><td>3. $a_n = C \cdot 2^n$</td></tr>
<tr><td>4. $a_n = C \cdot 3^n$</td></tr>
<tr><td>5. $a_n = 10 \cdot (2^n + 3^n)$</td></tr>
</table>

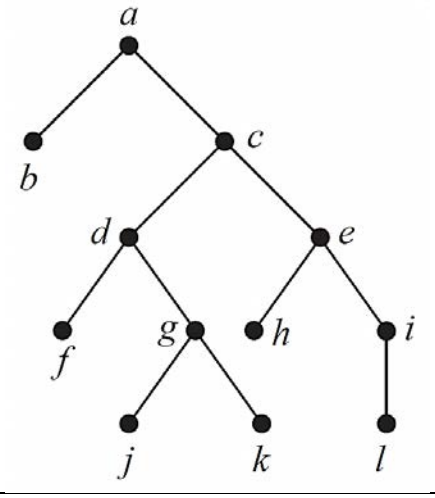
<table>
<tr>
<td>Assignment7
Write down all internal vertices of the rooted tree in the figure:
 
</td>
<td>1. d, f, g, h, j
2. b, c, i
3. a, b
4. d, f, g
5. c, d, e, i, g</td>
</tr>
<tr>
<td>Assignment 8
Use the coding scheme
<table><tr><th>Символ</th><th>Код</th></tr><tr><td>a</td><td>111</td></tr><tr><td>c</td><td>101</td></tr><tr><td>e</td><td>01</td></tr><tr><td>i</td><td>11011</td></tr><tr><td>g</td><td>0010</td></tr><tr><td>h</td><td>1001</td></tr><tr><td>m</td><td>0001</td></tr><tr><td>s</td><td>10001</td></tr><tr><td>t</td><td>11001</td></tr></table>to encode the word $agate$.</td>
<td>1. 111101101111001100010011
2. 11100101111100101
3. 1010101110010101010101
4. 1010111001010001010
5. 000111100010001111110011 0</td>
</tr>
<tr>
<td>Assignment 9
Find a regular expression corresponding to the regular set $\{ab, ac, ad, bb, bc, bd\}$.</td>
<td>1. $(a \vee b)(b \vee c)$
2. $a(\lambda \vee c \vee d)b$
3. $(a \vee b \vee c)(d \vee e \vee \mathrm{f})$
4. $(b \vee c \vee d)a$
5. $(a \vee b)(b \vee c \vee d)$</td>
</tr>
<tr>
<td>Assignment 10
Which of the following words are accepted by the following automaton?</td>
<td>1. $bbba$
2. aaa
3. $abbba$
4. $abaabab$
5. $abbbbaa$</td>
</tr>
</table>



Assignment 11

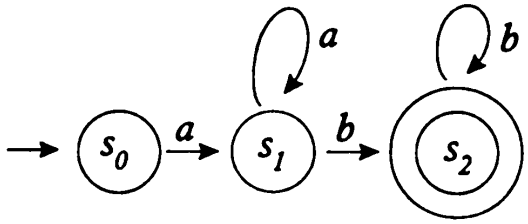
Representations of Boolean functions as DNF and as CNF.

<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the converse of this conditional statement. I come to class whenever there is going to be a quiz.</td><td>1. If I do not come to class, then there will be a quiz.</td></tr>
<tr><td>2. If I come to class, then there will be a quiz.</td></tr>
<tr><td>3. If I come to class, then there will not be a quiz.</td></tr>
<tr><td>4. If I do not come to class, then there will not be a quiz.</td></tr>
<tr><td>5. If there is not going to be a quiz, then I don’t come to class.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$ and $C = \{4, 5, 6, 7, 8, 9, 10\}$.
Find $A \cap B \cap C$.</td><td>1. {4, 6}</td></tr>
<tr><td>2. {0, 2, 4, 6}</td></tr>
<tr><td>3. {4, 5, 6}</td></tr>
<tr><td>4. {4, 5, 6, 7 , 8, 9, 10}</td></tr>
<tr><td>5. $\varnothing$</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of three-digit numbers made from the digits 6, 7, 8 and 9. How many numbers in S have no repeated digit?</td><td>1. 81</td></tr>
<tr><td>2. 4</td></tr>
<tr><td>3. 24</td></tr>
<tr><td>4. 1000</td></tr>
<tr><td>5. 27</td></tr>
<tr><td rowspan="5">Assignment 4
Find the sum-of-products expansion of the Boolean function $f(x, y, z)$, which is given in the table:
<table><tr><th>x</th><th>y</th><th>z</th><th>$f(x, y, z)$</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table></td><td>1. $x \wedge \bar{y} \wedge z$</td></tr>
<tr><td>2. $(\bar{x} \wedge y \wedge \bar{z}) \vee (x \wedge \bar{y} \wedge z)$</td></tr>
<tr><td>3. $\bar{x} \vee y \vee \bar{z}$</td></tr>
<tr><td>4. $x \vee \bar{y} \vee z$</td></tr>
<tr><td>5. $(x \wedge y \wedge z) \vee (x \wedge \bar{y} \wedge \bar{z})$</td></tr>
<tr><td rowspan="5">Assignment 5
Simplify the sum-of-products expansion of the Boolean function represented by the following K-map:
<table><tr><td></td><td>x</td><td>$\bar{x}$</td></tr><tr><td>y</td><td>1</td><td>1</td></tr><tr><td>$\bar{y}$</td><td>1</td><td></td></tr></table></td><td>1. $\bar{x} \vee \bar{y}$</td></tr>
<tr><td>2. y</td></tr>
<tr><td>3. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td>4. x</td></tr>
<tr><td>5. $x \vee y$</td></tr>
</table>

<table>
<tr><td>Assignment 6
Let A $= \{a, b\}$. Which of the following relations on a set A are symmetric?
$S = \{(a,a),(a,b),(b,a)\}$,
$T = \{(a,a),(a,b),(b,b)\}$,
$U = \{(a,b),(b,a)\}$,
$V = \{(b,b),(a,b),(b,a)\}$.</td><td>1. S, T, U, V
2. S, U, V
3. T
4. S, T
5. T, U</td></tr>
<tr><td>Assignment 7
Write down the left child of vertex c in the binary tree in the figure:
 
</td><td>1. a
2. d
3. e
4. f
5. g</td></tr>
<tr><td>Assignment 8
Use the coding scheme
<table><tr><th>Символ</th><th>Код</th></tr><tr><td>a</td><td>111</td></tr><tr><td>c</td><td>101</td></tr><tr><td>e</td><td>01</td></tr><tr><td>i</td><td>11011</td></tr><tr><td>g</td><td>0010</td></tr><tr><td>h</td><td>1001</td></tr><tr><td>m</td><td>0001</td></tr><tr><td>s</td><td>10001</td></tr><tr><td>t</td><td>11001</td></tr></table>to decode the word 11001100101000101.</td><td>1. $mathematics$
2. $gates$
3. $seismic$
4. $theme$
5. $agate$</td></tr>
<tr><td>Assignment 9
Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two bs.</td><td>1. $a^*ba^*ba^*$
2. $b^*ab^*ab^*$
3. $(a \cup c)^*b(a \cup c)^*b(a \cup c)^*$
4. $(b \cup c)^*a(b \cup c)^*a$
5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a$</td></tr>
</table>

Assignment 10

Which of the following words are not accepted by the following automaton?

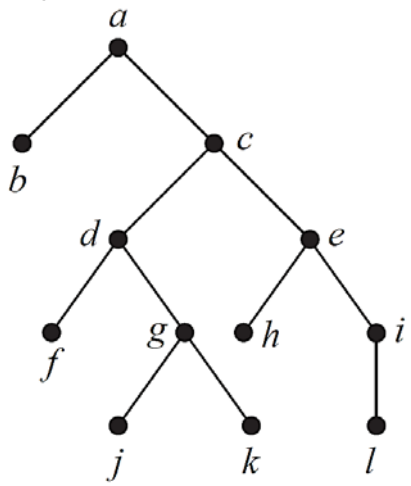


1. aab
2. $aabb$
3. abb
4. $aabbb$
5. aa

Assignment 11

Sets of functionally complete Boolean functions.

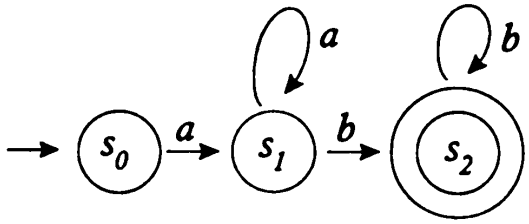
<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td rowspan="5">Assignment 1
State the contrapositive of this conditional statement. I come to class whenever there is going to be a quiz.</td><td>1. If I do not come to class, then there will be a quiz.</td></tr>
<tr><td>2. If I come to class, then there will be a quiz.</td></tr>
<tr><td>3. If I come to class, then there will not be a quiz.</td></tr>
<tr><td>4. If I do not come to class, then there will not be a quiz.</td></tr>
<tr><td>5. If there is not going to be a quiz, then I don’t come to class.</td></tr>
<tr><td rowspan="5">Assignment 2
Let $A = \{0, 2, 4\}, B = \{0, 1, 2, 3\}$ and $C = \{4, 5\}$. Find $A \cup B \cup C$.</td><td>1. {0, 1, 2, 3, 4, 5}</td></tr>
<tr><td>2. {0, 2, 4, 5}</td></tr>
<tr><td>3. {0, 1, 2, 3, 4}</td></tr>
<tr><td>4. {0, 1, 2, 3, 5}</td></tr>
<tr><td>5. $\varnothing$</td></tr>
<tr><td rowspan="5">Assignment 3
Let S denote the set of three-digit numbers made from the digits 1, 6, 7, 8 and 9. How many numbers in S have no repeated digit?</td><td>1. 5</td></tr>
<tr><td>2. 125</td></tr>
<tr><td>3. 1000</td></tr>
<tr><td>4. 75</td></tr>
<tr><td>5. 60</td></tr>
<tr><td rowspan="5">Assignment 4
Find the sum-of-products expansion of the Boolean function $f(x, y, z)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>z</th><th>$f(x, y, z)$</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table></td><td>1. $x \wedge y \wedge z$</td></tr>
<tr><td>2. $(\bar{x} \wedge y \wedge \bar{z}) \vee (x \wedge \bar{y} \wedge z)$</td></tr>
<tr><td>3. $\bar{x} \vee y \vee \bar{z}$</td></tr>
<tr><td>4. $x \vee \bar{y} \vee \bar{z}$</td></tr>
<tr><td>5. $(x \wedge y \wedge z) \vee (x \wedge \bar{y} \wedge \bar{z})$</td></tr>
<tr><td rowspan="5">Assignment 5
Simplify the sum-of-products expansion of the Boolean function represented by the following K-map:
<table><tr><th></th><th>x</th><th>$\bar{x}$</th></tr><tr><td>y</td><td>1</td><td>1</td></tr><tr><td>$\bar{y}$</td><td></td><td>1</td></tr></table></td><td>1. $\bar{x} \vee y$</td></tr>
<tr><td>2. y</td></tr>
<tr><td>3. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td>4. x</td></tr>
<tr><td>5. $x \vee y$</td></tr>
</table>

Assignment 6 The characteristic equation of a linear homogeneous recurrence relation with constant coefficients (for a sequence $\{a_n\}$) is of the form: $(r-5)\cdot(r-2)$. What is the general solution of this recurrence relation?	1. $a_n = C_1 \cdot 2^n + C_2 \cdot 5^n$ 2. $a_n = C \cdot 10^n$ 3. $a_n = C \cdot 2^n$ 4. $a_n = C \cdot 5^n$ 5. $a_n = 10 \cdot (2^n + 5^n)$
Assignment 7 Write down the right child of vertex c in the following binary tree:  	1. a 2. d 3. e 4. f 5. g
Assignment 8 Use the coding scheme to decode the word 11100101111100101.	1. $mathematics$ 2. $gates$ 3. $seismic$ 4. $theme$ 5. $agate$
Assignment 9 Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two as.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b$ 4. $(b \cup c)^*a(b \cup c)^*a(b \cup c)^*$ 5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a(a \cup b \cup c)^*$

Символ	Код
a	111
c	101
e	01
i	11011
g	0010
h	1001
m	0001
s	10001
t	11001

Assignment 10

Which of the following words are accepted by the following automaton?

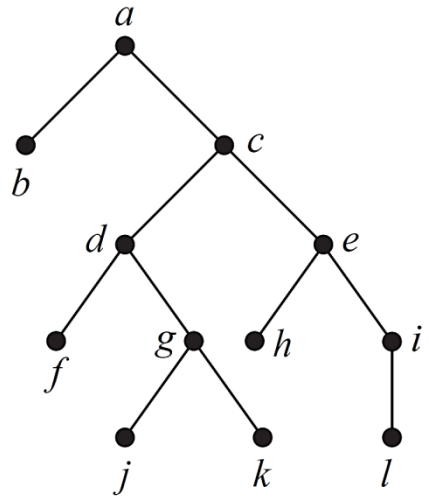
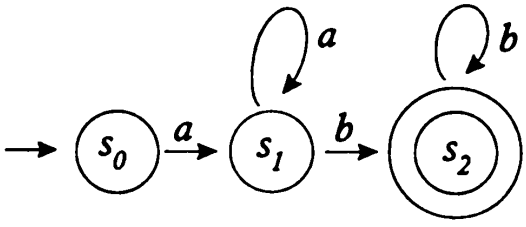


1. aa
2. aaa
3. abb
4. $aaaa$
5. ba

Assignment 11

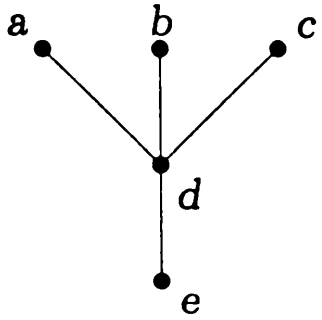
Minimization of a sum-of-products expansion. Karnaugh maps.

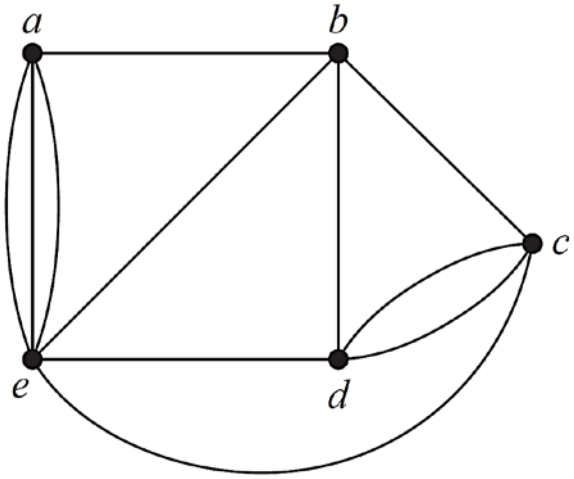
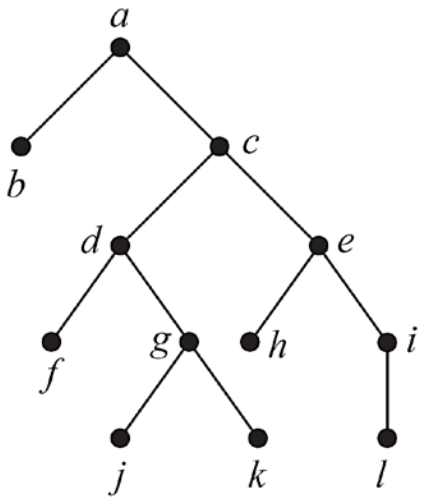
<table>
<tr><th>Test assignments</th><th>Answers</th></tr>
<tr><td>Assignment 1
Suppose that the domain of the propositional function $P(x)$ consists of the integers 1, 2 and 3. Write out the proposition $\exists x P(x)$ using conjunctions ($\wedge$), disjunctions ($\vee$) and negations ($^{-}$).</td><td>1. $P(1) \vee P(2) \vee P(3)$</td></tr>
<tr><td></td><td>2. $(P(1) \wedge P(2)) \vee P(3)$</td></tr>
<tr><td></td><td>3. $P(1) \wedge P(2) \wedge P(3)$</td></tr>
<tr><td></td><td>4. $\overline{P(1)} \wedge \overline{P(2)} \wedge \overline{P(3)}$</td></tr>
<tr><td></td><td>5. $\overline{P(1)} \vee \overline{P(2)} \vee \overline{P(3)}$</td></tr>
<tr><td>Assignment 2
Let $A = \{a\}$ and $B = \{y, z\}$. Find $A \times B$.</td><td>1. $(a, y), (a, z)$</td></tr>
<tr><td></td><td>2. a, y, z</td></tr>
<tr><td></td><td>3. $(a, a), (y, y), (z, z)$</td></tr>
<tr><td></td><td>4. $(a, y), z$</td></tr>
<tr><td></td><td>5. $(a, a), (y, z), (z, y)$</td></tr>
<tr><td>Assignment 3
Find the coefficient of $a^3 b^5$ in the expansion $(a + b)^8$.</td><td>1. 78</td></tr>
<tr><td></td><td>2. 56</td></tr>
<tr><td></td><td>3. 8</td></tr>
<tr><td></td><td>4. 1</td></tr>
<tr><td></td><td>5. 4</td></tr>
<tr><td>Assignment 4
Find the product-of-sums expansion of the Boolean function $f(x, y, z)$ which is given in the table:
<table><tr><th>x</th><th>y</th><th>z</th><th>$f(x, y, z)$</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table></td><td>1. $x \vee \bar{y} \vee z$</td></tr>
<tr><td></td><td>2. $(\bar{x} \wedge y \wedge \bar{z}) \vee (x \wedge \bar{y} \wedge z)$</td></tr>
<tr><td></td><td>3. $\bar{x} \vee y \vee \bar{z}$</td></tr>
<tr><td></td><td>4. $(x \vee \bar{y} \vee z) \wedge (\bar{x} \vee y \vee \bar{z})$</td></tr>
<tr><td></td><td>5. $(\bar{x} \vee y \vee z) \wedge (\bar{x} \vee \bar{y} \vee \bar{z})$</td></tr>
<tr><td>Assignment 5
Simplify the sum-of-products expansion of the Boolean function represented by the following K-map:
<table><tr><td></td><td>x</td><td>$\bar{x}$</td></tr><tr><td>y</td><td></td><td>1</td></tr><tr><td>$\bar{y}$</td><td>1</td><td>1</td></tr></table></td><td>1. $\bar{x} \vee \bar{y}$</td></tr>
<tr><td></td><td>2. y</td></tr>
<tr><td></td><td>3. $\bar{x} \wedge \bar{y}$</td></tr>
<tr><td></td><td>4. x</td></tr>
<tr><td></td><td>5. $x \vee y$</td></tr>
<tr><td>Assignment 6
Let A $= \{a, b\}$. Which of the following</td><td>1. S, T, U, V</td></tr>
<tr><td></td><td>2. S, U, V</td></tr>
</table>

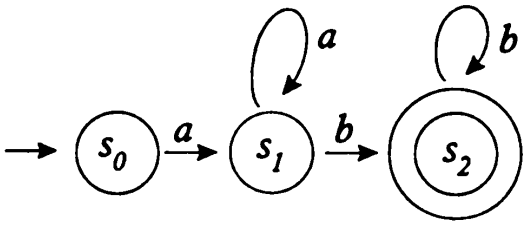
relations on a set A are reflexive and transitive? $S = \{(a,a),(a,b),(b,a)\}$, $T = \{(a,a),(a,b),(b,b)\}$, $U = \{(a,b),(b,a)\}$, $V = \{(b,b),(a,b),(b,a)\}$.	3. T 4. S 5. T, U
Assignment 7 Determine the depth of the rooted tree in the figure:  	1. 2 2. 3 3. 4 4. 5 5. 11
Assignment 8 An encoding function is defined by the generator matrix $G = \begin{bmatrix} 1\,0\,0\,1\,1\,1 \\ 0\,1\,0\,1\,0\,1 \\ 0\,0\,1\,1\,1\,1 \end{bmatrix}$. Encode 011.	1. 011010 2. 100111 3. 011101 4. 100110 5. 111100
Assignment 9 Let $A = \{a,b,c\}$. Give a regular expression for the set of all elements of A^*, containing exactly two bs.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b(a \cup c)^*$ 4. $(b \cup c)^*a(b \cup c)^*a(b \cup c)^*$ 5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a(a \cup b \cup c)^*$
Assignment 10 Find a regular expression for the language accepted by the automaton:  	1. $a^*ba^*ba^*$ 2. $ab(aab)^*$ 3. aa^*bb^* 4. a^*ba^* 5. $a^*ba^*ba^*b$

Assignment 11

Simplification of a sum-of-products expansion. The Quine-McCluskey method.

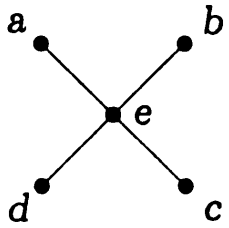
Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x, y)$ consists of pairs (x, y), where $x = 1, 2$ and $y = 0, 2, 3$. Write out the proposition $\forall x P(x, 3)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(1,3) \vee P(2,3)$ 2. $P(1,3) \wedge P(2,3)$ 3. $P(2,2) \vee P(2,3)$ 4. $P(2,0) \vee P(2,3)$ 5. $P(2,0) \wedge P(2,2)$
Assignment 2 List all elements incomparable with d in the partial ordering with the accompanying Hasse diagram.  	1. a, b, c 2. e 3. d 4. a, b, c, e 5. Such elements don't exist.
Assignment 3 A sequence of integers is defined by $f(0) = 1$, $f(1) = 3, f(k) = 2 \cdot f(k-1) - f(k-2)$. Calculate $f(3)$.	1. 1 2. 3 3. 7 4. 16 5. 5
Assignment 4 The Boolean operator $\oplus$, called the XOR operator, is defined by $x \oplus y = (\bar{x} \wedge y) \vee (x \wedge \bar{y})$. Simplify the expression $x \oplus x$.	1. 0 2. 1 3. x 4. $\bar{x}$ 5. $x \oplus x$
Assignment 5 Simplify the Boolean function $f(x, y, z, w) = (x \wedge y \wedge z \wedge w) \vee (\bar{x} \wedge y \wedge z \wedge w)$.	1. $y \wedge z \wedge w$ 2. $x \wedge z \wedge w$ 3. $x \wedge y \wedge w$ 4. $x \wedge y \wedge z$ 5. $\bar{x} \wedge y \wedge z \wedge w$
Assignment 6 Find an Eulerian path in the Eulerian graph in the figure:	1. $edbae$ 2. $edbe$ 3. $edcdbe$ 4. $edbaeae$ 5. $edbaeaebcdce$

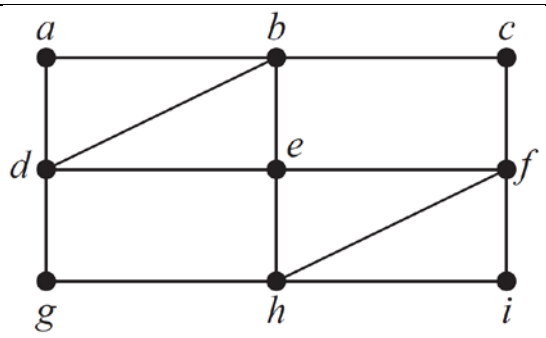
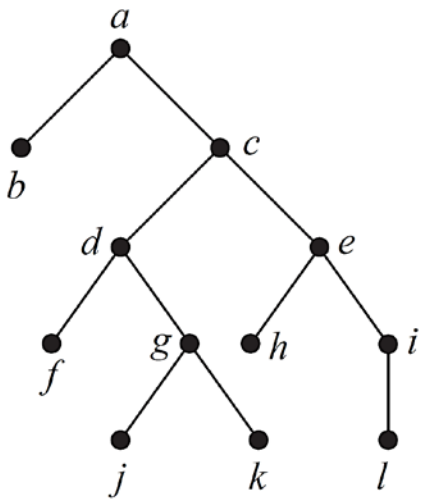
	
Assignment 7 Write down all leaves in the following binary rooted tree: 	1. e, g, h, i, j, k 2. b, c 3. b, f, j, k, h, l 4. d, e, f, g 5. e, g, h, i, j, k
Assignment 8 The minimum distance of the code is 3. Find the maximum number of errors which the code can detect.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Find a regular expression corresponding to the regular set $\{ba, ca, da\}$.	1. $(a \vee b)(b \vee c)$ 2. $a(\lambda \vee c \vee d)b$ 3. $(a \vee b \vee c)(d \vee e \vee f)$ 4. $(b \vee c \vee d)a$ 5. $a(b \vee c \vee d)$
Assignment 10 Find a regular expression for the language accepted by the automaton:	1. $a^*ba^*ba^*$ 2. $ab(aab)^*$ 3. aa^*bb^* 4. a^*ba^* 5. $a^*ba^*ba^*b$

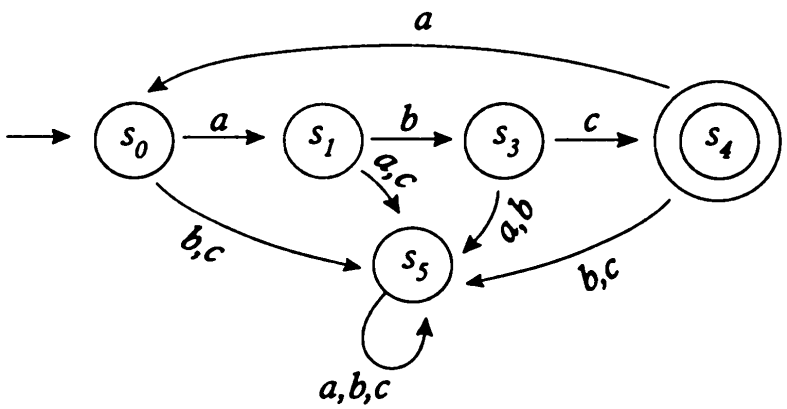


Assignment 11

Nondeterministic finite automata.

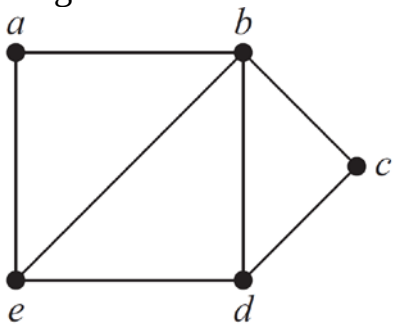
Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x, y)$ consists of pairs (x, y), where $x = 1, 2$ and $y = 0, 2, 3$. Write out the proposition $\exists x P(x, 3)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(1,3) \vee P(2,3)$ 2. $P(1,3) \wedge P(2,3)$ 3. $P(2,2) \vee P(2,3)$ 4. $P(2,0) \vee P(2,3)$ 5. $P(2,0) \wedge P(2,2)$
Assignment 2 List all elements incomparable with a in the partial ordering with the accompanying Hasse diagram.  	1. e, b, c 2. b 3. d, c 4. c, d, b 5. Such elements don't exist.
Assignment 3 A sequence of integers is defined by $f(0) = 2$, $f(1) = 4, f(k) = 3 \cdot f(k-1) - 2 \cdot f(k-2)$. Calculate $f(3)$.	1. 24 2. 8 3. 16 4. 4 5. 6
Assignment 4 The Boolean operator $\oplus$, called the XOR operator, is defined by $x \oplus y = (\bar{x} \wedge y) \vee (x \wedge \bar{y})$. Simplify the expression $x \oplus \bar{x}$.	1. 0 2. 1 3. x 4. $\bar{x}$ 5. $x \oplus \bar{x}$
Assignment 5 Simplify the Boolean function $f(x, y, z, w) = (x \wedge y \wedge z \wedge w) \vee (x \wedge \bar{y} \wedge z \wedge w)$.	1. $y \wedge z \wedge w$ 2. $x \wedge z \wedge w$ 3. $x \wedge y \wedge w$ 4. $x \wedge y \wedge z$ 5. $x \wedge \bar{y} \wedge z \wedge w$
Assignment 6 Find an Eulerian path in the Eulerian graph in the figure:	1. $fihefhef$ 2. $fedbehfcbadghif$ 3. $fcbadghif$ 4. $fedbehf$ 5. $fcbef$

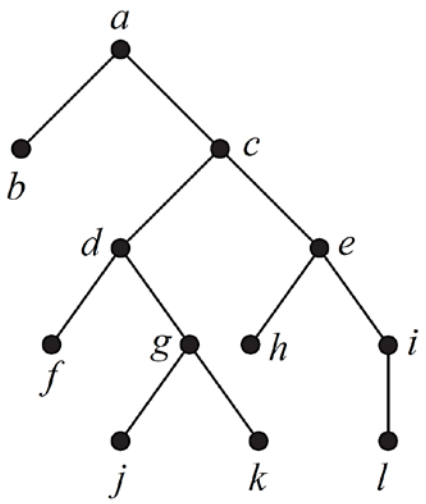
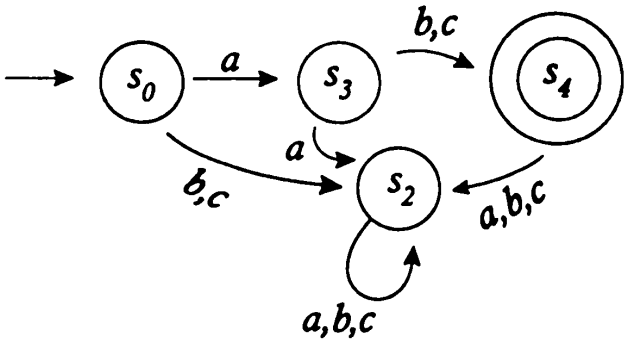
	
Assignment 7 Write down all internal vertices of the following rooted tree 	1. d, f, g, h, j, k 2. b, c, e, i, d, g 3. c, e, i, d, g 4. d, f, g, j, k 5. b, c, d, f
Assignment 8 The minimum distance of the code is 3. Find the maximum number of errors which the code can correct.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Find a regular expression corresponding to the regular set $\{ab, ac, ad, bb, bc, bd\}$.	1. $(a \vee b)(b \vee c)$ 2. $a(\lambda \vee c \vee d)b$ 3. $(a \vee b \vee c)(d \vee e \vee f)$ 4. $(b \vee c \vee d)a$ 5. $(a \vee b)(b \vee c \vee d)$
Assignment 10 Find a regular expression for the language accepted by the automaton:	1. $a^*ba^*ba^*$ 2. $abc(aabc)^*$ 3. $a(b \vee c)$ 4. a^*ba^*c 5. $a^*ba^*ba^*c$

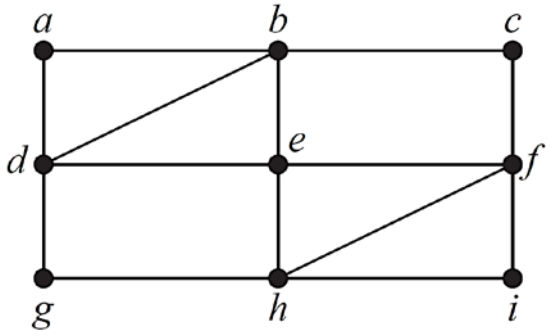


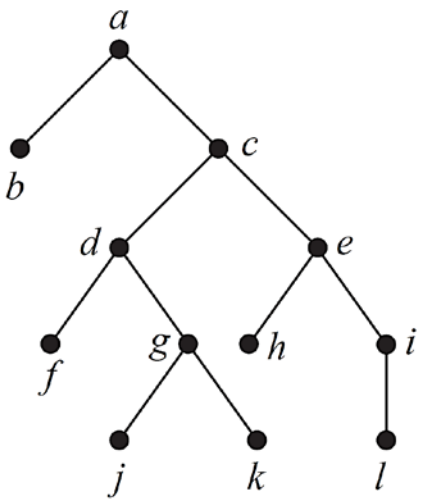
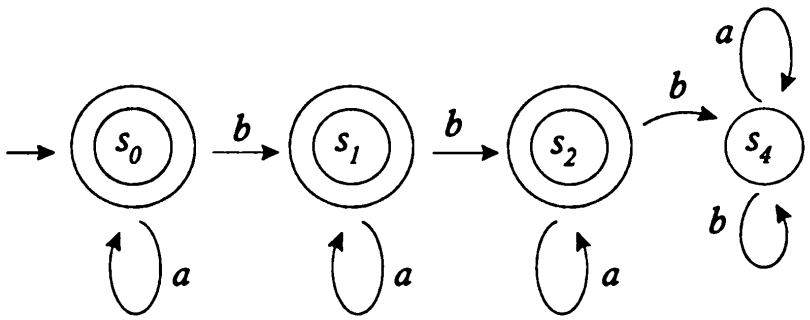
Assignment 11

Deterministic finite automata.

Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x,y)$ consists of pairs (x,y), where $x=1,2$ and $y=0,2,3$. Write out the proposition $\forall y P(2,y)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(2,0)\vee P(2,2)\vee P(2,3)$ 2. $P(2,0)\wedge P(2,2)\wedge P(2,3)$ 3. $P(2,2)\vee P(2,3)$ 4. $P(2,0)\vee P(2,3)$ 5. $P(2,0)\wedge P(2,2)$
Assignment 2 Suppose that $A=\{1,2,3,4,5,6,7,8,9,10\}$. Let R be the relation on the set A such that $(x,y)\in R$ if and only if $x-y$ is an even integer. R is an equivalence relation. What is the equivalence class [3] for this equivalence relation?	1. $\{2,4,6,8,10\}$ 2. $\{1,3,5,7,9\}$ 3. $\{6\}$ 4. $\{3,6,9\}$ 5. $\emptyset$
Assignment 3 Find the number of elements in the union $A_1\cup A_2\cup A_3$ if there are 100 elements in each set A_1,A_2,A_3 and these sets are pairwise disjoint.	1. 100 2. 300 3. 225 4. 200 5. 150
Assignment 4 The Boolean operator | , called the NAND operator, is defined by $x\mid y=\overline{x\wedge y}$. Express $\bar{x}$ using just the | operator.	1. $x\mid(x\mid x)$ 2. $\bar{x}\mid x$ 3. $x\mid x$ 4. $x\mid(\bar{x}\mid\bar{x})$ 5. $\bar{x}\mid(x\mid x)$
Assignment 5 Simplify the Boolean function $f(x,y,z,w)=(x\wedge y\wedge z\wedge w)\vee(x\wedge y\wedge\bar{z}\wedge w)$.	1. $y\wedge z\wedge w$ 2. $x\wedge z\wedge w$ 3. $x\wedge y\wedge w$ 4. $x\wedge y\wedge z$ 5. $x\wedge y\wedge\bar{z}\wedge w$
Assignment 6 Find a Hamiltonian cycle in the Hamiltonian graph in the figure:  	1. $ebcde$ 2. $ebcdbe$ 3. $abcdea$ 4. $ebde$ 5. $eabde$

Assignment 7 Write down the left child of vertex d in the following binary tree:  	1. b 2. c 3. e 4. f 5. g
Assignment 8 The minimum distance of the code is 4. Find the maximum number of errors which the code can detect.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two bs.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b(a \cup c)^*$ 4. $(b \cup c)^*a(b \cup c)^*a$ 5. $(a \cup b \cup c)^*a(a \cup b \cup c)^*a$
Assignment 10 Find a regular expression for the language accepted by the automaton:  	1. $a^*ba^*ba^*$ 2. $abc(aabc)^*$ 3. $a(b \vee c)$ 4. a^*ba^*c 5. $a^*ba^*ba^*c$
Assignment 11 Grammars.	

Test assignments	Answers
Assignment 1 Suppose the domain of the propositional function $P(x,y)$ consists of pairs (x,y), where $x=1,2$ and $y=0,2,3$. Write out the proposition $\exists y P(2,y)$ using conjunctions ($\wedge$) and disjunctions ($\vee$).	1. $P(2,0)\vee P(2,2)\vee P(2,3)$ 2. $P(2,0)\wedge P(2,2)\wedge P(2,3)$ 3. $P(2,2)\vee P(2,3)$ 4. $P(2,0)\vee P(2,3)$ 5. $P(2,0)\wedge P(2,2)$
Assignment 2 Suppose that $A=\{1,2,3,4,5,6,7,8,9,10\}$. Let R be the relation on the set A such that $(x,y)\in R$ if and only if $x-y$ is an even integer. R is an equivalence relation. What is the equivalence class $[6]$ for this equivalence relation?	1. $\{2,4,6,8,10\}$ 2. $\{1,3,5,7,9\}$ 3. $\{6\}$ 4. $\{3,6,9\}$ 5. $\emptyset$
Assignment 3 There are x computer science students at a university. Of these, 94 have taken a course in $C^{\#}$, 70 have taken a course in C^{++} and 81 have taken a course in $Java$. Each computer science student has taken a course only in one programming language. Find x.	1. 151 2. 175 3. 245 4. 164 5. 140
Assignment 4 The Boolean operator $\downarrow$, called the NOR operator, is defined by $x\downarrow y=\overline{x\vee y}$. Express $\bar{x}$ using just the $\downarrow$ operator.	1. $x\downarrow(x\downarrow x)$ 2. $\bar{x}\downarrow x$ 3. $x\downarrow x$ 4. $x\downarrow(\bar{x}\downarrow\bar{x})$ 5. $\bar{x}\downarrow(x\downarrow x)$
Assignment 5 Simplify the Boolean function $f(x,y,z,w)=(x\wedge y\wedge z\wedge w)\vee(x\wedge y\wedge z\wedge\bar{w})$.	1. $y\wedge z\wedge w$ 2. $x\wedge z\wedge w$ 3. $x\wedge y\wedge w$ 4. $x\wedge y\wedge z$ 5. $x\wedge y\wedge z\wedge\bar{w}$
Assignment 6 Find a Hamiltonian cycle in the Hamiltonian graph in the figure: 	1. $ecbe$ 2. $ebadghifce$ 3. $ecbae$ 4. $ecbaeighe$ 5. $efcbade$

Assignment 7 Write down the right child of vertex d in the following binary tree:  	1. a 2. c 3. e 4. f 5. g
Assignment 8 The minimum distance of the code is 5. Find the maximum number of errors which the code can correct.	1. 0 2. 1 3. 2 4. 3 5. 4
Assignment 9 Let $A = \{a, b\}$. Give a regular expression for the set of all elements of A^*, containing exactly two as.	1. $a^*ba^*ba^*$ 2. $b^*ab^*ab^*$ 3. $(a \cup c)^*b(a \cup c)^*b$ 4. $(b \cup c)^*a(b \cup c)^*a(b \cup c)^*$ 5. $(a \cup b \cup c)^*aa(a \cup b \cup c)^*$
Assignment 10 Find a regular expression for the language accepted by the automaton:  	1. $a^*ba^*ba^*$ 2. $abc(aabc)^*$ 3. $a(b \vee c)$ 4. a^*ba^*c 5. $a^*ba^*ba^*c$
Assignment 11 Turing machines.	